Intelligent Hospital 2 (2026) 100043

Contents lists available at ScienceDirect

Intelligent Hospital

journal homepage: www.keaipublishing.com/en/journals/intelligent-hospital/

Review

Digital twin technologies in medicine: The innovations, barriers, and future directions

Gowtham Menon *, Bhagyashri Malave, Mayuri Mhaske, Sakshi Parjane, Hrushikesh Mhaismale

Department of Pharmaceutics, Sanjivani College of Pharmaceutical Education & Research, Savitribai Phule Pune University, Kopargaon, Maharashtra -423603, India

ARTICLE INFO

Keywords:
Digital twin
Healthcare
Artificial intelligence
Machine learning
Internet of medical things (IoMT)
Big data
Cloud computing
Personalized medicine

ABSTRACT

Digital Twin technology revolutionizes modern medicine with the creation of real-time virtual models of patients, organs, and healthcare systems; all of these improve diagnosis, treatment, and management. From industrial applications, DTs moved to the healthcare sector driven by advancements in AI, IoMT, big data analytics, and cloud computing. This review synthesizes recent findings from systematic literature reviews across databases such as PubMed, Scopus, and Web of Science, focusing on key themes including personalized medicine, predictive modeling, virtual clinical trials, and optimization of healthcare systems. Such applications are performed with the intention of minimizing risks within treatments and enhancing patients' experiences. In contrast, substantial obstacles to this goal include data privacy, systemic incompatibility, cyber threats, and regulatory problems that all act to limit the wide diffusion of these technologies. Overcoming these challenges will require ethical governance, standardized systems, and interdisciplinary collaboration for full empowerment of DTs. Ongoing technological advancements strive to create a healthcare environment that is efficient, predictive, and focused on patients' experiences, exemplified by the capability to detect early warnings from subtle indicators in physiological monitoring.

1. Introduction

Digital Twins (DTs) are advanced, dynamic virtual replicas of physical objects, including a specific patient, a specific organ, or the entire healthcare system. The latest sources of information used to continually optimize such copies include wearable devices, electronic health records, medical imaging, genetic data, and biosensors.[1,2] DTs will allow personalized medicine by enabling the prediction of treatment response, monitoring disease progression, and informing clinical decisions by perfectly mimicking biological processes and health states.[3,4] They are artificially intelligent and data-driven, converting voluminous datasets into practical knowledge.[5] Although originally designed to support industries such as aerospace, modern digital twins have become essential in research and precision medicine and operational enhancement in the healthcare industry.[6]

The current development of digital technologies has dramatically increased the adoption of digital twin (DT) systems into the healthcare sector, turning conventional medical practice into a smart and data-driven model. Artificial Intelligence (AI), the Internet of Medical Things (IoMT) and real-time big data analytics have come together to form dynamic and patient-centered virtual models, which can simulate physiological processes and predict the evolution of the disease with great precision. By means of these innovations, clinicians can evaluate patient data in real-time, predict clinical risks, and create personal treatment interventions on the basis of digital simulations. For example, IoT and deep learning have demonstrated impressive capabilities in the early detection of diseases, monitoring pandemics and high-scale health surveillance where sensor-collected data are processed within seconds to assist clinicians in their decision-making.[7] Likewise, mobile edge computing architectures are being designed to streamline data flow and minimize latency in DT applications to enable faster processing of medical data at the network edge, where only a centralized cloud system is not the sole solution.[8] Such developments play a vital role in healthcare processes that require timeliness like remote monitoring and emergency diagnostics.

Recent bibliometric and systematic reviews have found that there is a

Peer review under responsibility of Editorial Board of Intelligent Hospital.
* Corresponding author.
E-mail address: gowthammphd@gmail.com (G. Menon).

https://doi.org/10.1016/j.inhs.2025.100043
Received 19 September 2025; Received in revised form 11 November 2025; Accepted 11 November 2025
Available online 16 November 2025
3050-8371/© 2026 The Authors. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

growing research focus on DT-enabled personalized medicine, organ-level modeling and AI-powered health prediction systems, which all play a role in more precise and adaptive healthcare approaches.[9] Moreover, topic modeling of DT literature has demonstrated the new research trends that are aimed at predictive diagnostics, virtual clinical trials, and optimization of healthcare systems indicating the transition between the conceptual frameworks to real-life application in hospitals and research organizations.[10] The significance of creating reliable, interoperable, and scalable DT architectures to protect patient data privacy, enable harmonious data exchange among clinical systems and provide real-time synchronization between physical and virtual models has also been complemented by complementary technological studies.[11]

Bruynseels et al.[2] introduced the concept of digital twins (DTs) in healthcare, demonstrating how they can generate real-time virtual replicas of patients, organs, and systems while also highlighting ethical and data-integration challenges.[12] Scientists examined applications of DT in precision medicine and emphasized that heterogeneous data fusion and strict validation are necessary to support clinical decisions. Subsequent empirical studies, in acute care, provided evidence of successful DT implementation, with a usability score indicating acceptance by clinicians.[13] The comparative analyses showed that improvements in accuracy, speed, and cost-efficiency were achieved through DT-augmented interventions compared to conventional approaches.[14] Ringeval et al.[13] later conducted a meta-review of 25 studies and emphasized that, although DTs are ideal for personalized care and operational modeling, significant gaps remain in scalability, interoperability, multi-center validation, and real-world implementation.[15] These ongoing challenges highlight the need for systematic synthesis of existing DT innovations, performance benchmarking, and practical frameworks. Inspired by this landscape, the current study will critically examine recent DT technologies in the clinical field, evaluate experimental findings and performance comparisons, identify key barriers, and propose a strategic agenda to accelerate their integration into standard clinical practice.[16]

1.1. Background and objectives

Digital Twin (DT) technology is a system and that originally targeted industrial and engineering systems and has recently attracted a high level of attention in healthcare due to its capability to generate real-time, data-driven virtual models of patients, organs, or healthcare systems. Although this has generated increasing attention, existing literature is disjointed and frequently fails to consider the synergies of Artificial Intelligence (AI), the Internet of Medical Things (IoMT), big data analytics, and cloud computing, which are essential in the clinical translation. Moreover, issues related to interoperability of data, privacy and security, the ethical regulation, and standardization are not properly tackled.

The rationale behind this review is thus the necessity to summarize the existing literature and critically evaluate it, to determine the main technological and ethical obstacles, and map the future research opportunities to inform the successful implementation of DT technologies in contemporary medicine. The main purpose of the review is to present an overall and critical synthesis of the existing trends in digital twin applications in healthcare, identifying their technological backgrounds, their integration with AI and data-driven tools, clinical use, the major obstacles in its implementation, and future research directions to support the effective, ethical, and standard use of DT technology in contemporary medical practice.

The primary goals of this review are, therefore, as follows:

- To deliver a general review of DT technologies in healthcare, it is important to focus on its conceptual basis and combine it with AI, IoMT, and data analytics.
- To summarize and critically review the recent developments and uses of DTs in personalized medicine, predictive modeling, virtual clinical trials, and optimization of health care systems.
- To address and discuss the most significant barriers and challenges to the implementation of DTs, such as interoperability, data privacy, ethics, regulation, and cybersecurity.
- To present future trends and possibilities to improve the adoption of DT using the standardized frameworks, ethical governance, and cross-disciplinary cooperation.

1.2. Organization of the Study

The rest of the manuscript is organized as follows: 2 describes the origins and history of digital twin technology throughout the industrial sphere until its use in health care. 3 outlines the methodology that was used to select and synthesize the literature. 4 describes the digital twins use in modern healthcare systems. 5 covers the concept of digital twins in healthcare, the essential elements, types of digital twins in healthcare cyber-physical systems in digital technology. 6 covers the topic of enabling technologies, such as the Internet of Medical Things (IoMT), artificial intelligence (AI), and cloud computing. 7 examines interoperability and data standards. 8 explores the applications of digital twins, and 9 & 10 deals with the benefits, challenges, and limitations. 11 gives the future opportunities. 12 covers the conclusion.

2. Origins and history of digital twins: industry to medicine

Digital twins have their roots in industrial engineering, where the need to monitor, optimize, and forecast the maintenance demands of complex systems led to the development of virtual models representing physical assets.[17] NASA used an initial version of such a concept during the Apollo space program in the 1960s.[18,19] The NASA engineers also came up with computer-based versions of spacecraft components that could be simulated on the ground in real-time to help them debug problems that occurred during flights.[20,21] Early uses of the concept did not employ the term digital twin, but they established the principles of system duplication and simulation on which the use of the practice today is based.[22] The official term digital twin, was coined by Dr. Michael Grieves of the University of Michigan in 2002 when applying the product lifecycle management (PLM) idea to setting up a manufacturing organization.[23,24]

The goal was to create a digital replica of a physical product that would define, anticipate, and enhance its performance throughout its lifecycle, from design and production to maintenance and eventual disposal. It was soon adopted by the manufacturing sector, particularly in industries requiring high precision, such as aerospace, automotive, and heavy machinery.[25,26] Digital twin technologies were implemented by companies like General Electric, Siemens, and Rolls-Royce to monitor real-time data on aircraft engines or factory operations and other machinery tools.[27,28]

The applications reduced operational costs, improved efficiency, and enhanced safety.[1,29] The rapid development of sensors, Internet of Things, cloud computing, and artificial intelligence drove the advancement of digital twins.[30] These technologies enable real-time data gathering, transmission, and analysis on large volumes, thereby driving the development of more dynamic and responsive digital twins. As digital twins become more powerful and widely accessible, their potential application beyond settings, particularly in healthcare-gained increasing attention. The digital twin technology migration to healthcare naturally corresponded as healthcare was becoming increasingly data-dependent. With the advent of electronic health records (EHRs), wearable health devices, imaging technologies, and genomic sequencing, healthcare professionals began generating vast amounts of data comparable to the high-order datasets observed in industrial systems.[31–34]

The ensuing data deluge was a fertile field to create and use medical

digital twins early on. First, medical uses of DTs were, to a great extent, organ specific. The Living One, one of the first major projects in the field was the Heart Project of Dassault Systems in 2013. In that effort it attempted to produce a computer simulation of the functioning of the human heart as useful as a real heart. Its digital duplicate could test the blood circulation, perform the virtual surgeries, and analyze the effects of medical equipment without endangering the real patients.[38] The early success of this technology demonstrated how digital twins could provide unprecedented understanding of human physiology and clinical outcomes. It was then followed by extending the idea to include patient-specific digital twins that integrated personal clinical data, heredity, and lifestyle into the modelling of disease pathology and response to treatment.[39] As an example, digital twins have been used to predict how cancer patients would respond to different chemotherapy regimens, or how the patient would respond to surgical operations in cardiovascular diseases.[40] Besides creating personalized care, digital twins have found use in system-related problem-solving within healthcare facilities. As a means of resolving the emergency during the COVID-19 pandemic, hospital executives started using digital twins to optimize the capacity management of a hospital and plan the effective allocation of resources in case of an emergency. Simulation models similar to DTs were also used in the pandemic to forecast ICU bed demands, ventilator usage, and personnel requirements.[41]

These examples point out the more generalized nature of digital twins and the capacity they possess to enable healthcare not only at the patient level, but also on a system-wide scale. The evolution of digital twins from industrial application to healthcare represents a highly significant paradigm shift. Where the previous drivers were mostly in the area of engineering systems operational efficiency, the medical field is in the stage of adopting DTs currently in order to have clinical understanding, personalized medicine, and optimal design and redesign of systems that follows in Fig. 1. The bedside transition is parallel to the larger transition to data-enabled, precision medicine and suggests the significance of multidisciplinary teaming of engineers, clinicians, data scientists, and policymakers.[42,43] As the digital twin technology advances, its impact on the healthcare realm is likely to grow in the future, with the potential to transform how diseases are diagnosed and treated. Improved interoperability and machine learning tools are among the features that may enable digital twin tools to become standard instruments in future healthcare systems planning.[44]

3. Methodology

This review was based on systematic use to review existing literature on Digital twin (DT) technology and its role in healthcare. Keywords like Digital Twin, Artificial Intelligence (AI), Internet of Medical Things (IoMT), and healthcare applications were used to search the major databases including PubMed, Scopus, and Web of Science. They included peer-reviewed articles, case studies, and technology reports up to 2025. Literature has been reviewed to emphasize progresses, applications in practice, challenges, and perspectives of where patient-specific, organ level or system level digital twin applications are applicable. The corresponding information was mined and combined to point out the current trends, benefits, weaknesses, and opportunities in the healthcare field.

4. Digital twins in modern healthcare systems

Digital twins' translation to medicine comes in the form of the ability to coordinate the loop between data and decisions. DTs present an unrivalled advantage in an environment in which the use of real-time information, individualized treatment, and maximization of outcomes is so crucial. DTs also make precision medicine possible by reproducing a patient-specific response to various treatment strategies, so that a clinician can choose the least risky and most effective treatment.[45] Within operating rooms, digital twins allow planning of a procedure beforehand and even calculate the risk by simulating specific patient anatomy with high precision. Besides one-on-one care, DTs are also involved in running the hospital and systems. They can simulate patient flow, resource use, and emergency readiness plans, and enable hospitals to streamline their processes, reduce expenses, and enhance service quality.[46–48]

Digital twins enable virtual clinical trials by simulating and aiding in the development of many new treatments, which would otherwise require months or years to develop not to mention animal or early-phase human experiments.[41]With the world's healthcare systems becoming more data-and outcomes-focused, DTs are providing a method of bringing high-volume and high-variety data into predictable, actionable

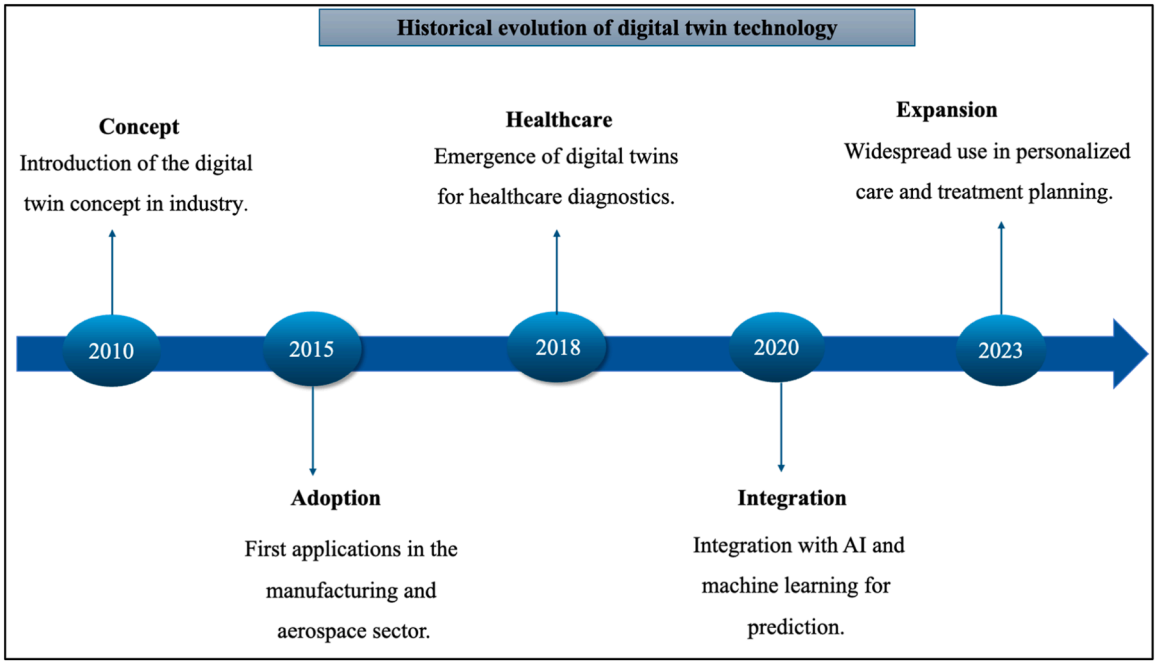


Fig. 1. Historical evolution of digital twin technology.[35–37]

models. They have been utilized in line with the worldwide healthcare trends such as personalized medicine, telemedicine, and AI-based diagnostics.[49,50]

5. Concept of digital twins in healthcare

The digital twin (DT) concept is based on the idea of incorporating physical systems and their digital representations into the same functional structure. Fundamentally, a healthcare digital twin is a real-time, continuously changing digital representation of a physical object, be it a patient, a given organ, or an entire healthcare facility.[51,52] They are real-time representations of models that can stimulate physiological processes, predict and assist clinical and operational decisions.[53]

5.1. Main elements of digital twins in healthcare

In healthcare, digital twins (DTs) consist of three basic and interconnected parts: the physical entity, the virtual model, and the real-time data link. These components operate in a closed circuit where it is possible to constantly synchronize the real-world counterpart and its digital representation.[54] This setup facilitates predictive analytics, personalized medicine, effective operations, and proactive healthcare interventions.[44,54]

5.1.1. Physical entity

The real-world representation of a digital twin is the physical entity and represents the basis on which the system is established.[44] In healthcare, this can be in the shape of a single patient, a particular organ, a piece of medical equipment, or even a healthcare institution. The physical entity of the patient incorporates demographic information, medical history, genetic information, and a continuously growing amount of clinical data collected at home (through wearable devices, with data sent via the Internet, as well as during visits to clinics).[55] At the organ level, organ twins, like cardiac, neurological, or pulmonary twins, recreate finer-scale structural and functional dynamics that can be used as diagnostic tools, treatment-planning tools and surgical simulators. Medical equipment twins, such as pacemakers, ventilators, or infusion pumps, enable predictive maintenance and performance monitoring, as the current functional data helps connect them to the virtual model. At a macro level, healthcare facility twins include workflows, resource utilization, and patient flow, all aimed at increasing operational efficiency. The main purpose of the physical entity (irrespective of its size) is to provide constant, high-quality, and context-informed data through sensors, connected devices, imaging systems, and clinical workflows, so that the digital twin serves as an accurate, living representation of the real system it model.[53,56]

5.1.2. Virtual model

In the virtual model, a computationally generated replica of the physical object that is intended to faithfully reflect the structure, the functioning and the behavior of the object is constructed.[53] It is deployed as a synthesis of mathematical modelling, machine learning algorithms, artificial intelligence techniques along with rich clinical datasets. Depending on scope and intended use, this model may be able to represent biological processes, operational workflows or the interaction between two or more systems.[49] As an example, a cardiac digital twin can be used to simulate the electrical conduction pathways in the heart, ventricular muscle contractions, and blood flow patterns, so that doctors can determine the risk of potential arrhythmias, plan interventions, and test valve replacement surgeries. In a comparable way, a neurological twin would be able to model brain connectivity dynamics, seizure initiation patterns, or neurodegenerative processes, over time, to inform patient-specific treatment approaches to epilepsy or Alzheimer's disease.[57,58] At the systems level, a hospital twin will be able to simulate the patterns of admission and release of patients, staffing efficiency, potential resource bottlenecks, and use of medical equipment. The time-sensitive simulations and what-if scenarios run by the virtual model enable the healthcare teams to estimate alternate strategies and approaches without interfering with patient care or hospital practice. Notably, the model is dynamic: its predictive performance and the ability to make decisions increase with time as it receives fresh information from the physical entity, and hence it is a dynamic, learning representation as opposed to a fixed digital replica.[59,60]

5.1.3. Real-time data latched

The most important data channel is the real-time link of communication between the physical and virtual representation, which allows the real-time continuous updating of the two models.[61] This connection ensures that the changes of the real-world system either physiological in a patient or operational in a hospital are captured immediately in the digital twin. The accumulation of data occurs with a variety of technologies, such as wearables (the heart rate, the breathing process, blood glucose levels or physical activity). Implants (pacemakers, neurostimulators), and the Internet of Medical Things (IoMT) platforms that integrate a number of medical devices into one network. Other means of inputs are Electronic Health Records (EHRs) that harbor longitudinal patient information; medical imaging systems like MRI, CT, and ultrasound scanners; and laboratory information systems that include data about biochemical, molecular, and genomic tests. The smooth flow of this information, of course, permits the virtual model to perform real-time modelling, predict adverse events before they happen, and prescribe interventions in a time-sensitive manner. As an example, the system may detect small changes in the heartbeat of a heart patients and automatically adjust medication or deliver a message to the physician. Deriving such data in real time (bed occupancy, emergency department inflow, and staff availability), as well as reallocation of resources in near real time to optimal status, is possible in a repeating this process, turning the digital twin into an active, decision-making assistance system that has the potential to elevate clinical outcomes, as well as minimize operational delays and hazards to patient safety.[62–66]

5.2. Types of digital twins in healthcare

Digital twins in healthcare may be divided into some categories according to the complexity, scale, and degree of application focus that follows in Fig. 2. All the types have specific purposes within the greater healthcare system:

5.2.1. Patient digital twins

Patient digital twins are very advanced models that reflect an individualized copy of the biological, physiological, and lifestyle data about a patient. To create an individualized virtual twin, they combine sets of different data on genetics, physiological measures, behavioral trends, and environmental factors. The main objective of these models is to facilitate precision and personalized healthcare, as they simulate the specific disease condition of a patient and forecast health outcomes with a very high degree of accuracy.[67,68] These twins also have the capability of predicting how conditions advance and progress, predicting how a given patient would respond to the different treatments, and streamlining clinical decisions about drug dosage and surgical timing to an optimal point. As an example, in the field of oncology, a digital twin of a patient may be created that recreates the growth patterns of tumors and models the outcome of various chemotherapy treatments.[69] This will enable clinicians to identify the most efficient course of treatment with the fewest potential side effects, facilitating a reduction in trial and error when choosing the treatment course, and resulting in more successful patient outcomes. Over time, such twins can evolve, incorporate new information, and remain an effective model of the patient's evolving health status.[70,71]

5.2.2. Digital twins at the organ level

Digital twins are specialized models based on the detailed

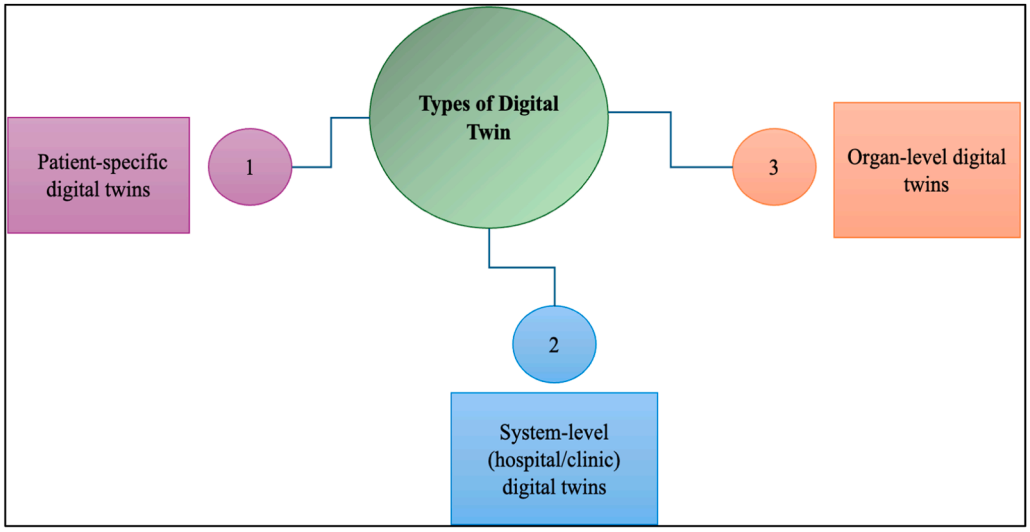


Fig. 2. Types of digital twins.[67,72]

organization and function of a single organ or physiological system. These models are extremely useful for clinical diagnostics, surgical planning, and medical education, allowing healthcare practitioners to study the organ-specific reactions without invasive tests. They can simulate the anatomical structure and physiological behavior of an organ under different conditions using advanced imaging, biomechanical modelling, and physiological data. One such model is the heart, which has been digitally modelled to test the electrical conduction system, blood flow dynamics, and mechanical contractions. Such models are invaluable in complex operations, such as valve replacements, ablation procedures for arrhythmias, or testing implantable devices before surgery. Similarly, centers like the King Abdullah Neuroscience Institute utilize brain digital twins to simulate neurological processes, predict responses to neurostimulation treatments. These complex organ-level simulations enable clinicians to make accurate, low-risk decisions, particularly in surgeries that demand maximum precision.[67,73]

5.2.3. *System-level digital twins (hospitals/ clinics)*

Digital twins at the system level extend the concepts beyond individual patients or organs to the dynamic operations of healthcare facilities in general.[45] These twins model the overall operations of clinics or hospitals, including the visualization of patient flow among departments, resource allocation, utilization of equipment, and staff scheduling. By integrating data collected from hospital information systems, IoT-connected medical devices, and operational records, these models provide administrators with actionable insights to improve efficiency and readiness.[74–77] System-level digital twins also assist administrators in anticipating peak admission periods, resource use, and contingency in the event of an emergency such as a pandemic or natural disaster.[78] One such situation was witnessed during the COVID-19 pandemic, where hospitals employed models based on simulation to forecast the requisite ICU beds, ventilators, and the number of staff. This made it possible to plan more accurately and better manage the supply chain, as well as respond quickly to variable patient volumes. In addition to crisis management, these digital twins can be used to continuously optimize the normal processes of the hospital to reduce wait times, increase patient throughput, and enhance overall quality of care.[41]

5.3. *Cyber-physical systems in digital twin technology*

They are the base infrastructure that facilitates an easy connection and communication between the near real world and the digital representation of the world. As a concept pertinent to the application of digital twins (DTs) in healthcare, Cyber-Physical Systems serve as a technological foundation of the continuous and dynamic digital exchanges of information between patients, clinical settings, and virtual models. They are technical combinations of computational algorithms, networking, and embedded hardware that ensure digital twins run in real time and dynamically reflect the current state of the physical system they model. In healthcare, CPS serve as a hub where information is gathered, manipulated, analysed, and applied to update digital twins. Their importance cannot be overstated, as the success of a digital twin is determined by the quality of the data it receives and uses to achieve the desired outcomes.[79,80]

5.3.1. *Data acquisition*

CPS can capture real-time physiological and environmental data from a large number of sources, including:

a) Health-related devices (wearables, e.g., smartwatches, fitness bands).
b) Implanted medical devices (e.g., pacemakers, continuous glucose monitors).
c) Clinical equipment (e.g., ECG machines, infusion pumps).
d) Health information systems and electronic medical imaging systems.
e) Electronic Health Records (EHRs).

These devices generate vast amounts of data related to vital signs, activity levels, medication adherence, and other clinically relevant aspects. CPS process and normalize this information to provide input to the virtual model, enabling the digital twin to recreate the current physiological state of a patient or the operational status of healthcare systems.[81,82]

5.3.2. *Real-time synchronization*

Digital twins depend on the significance of real-time synchronization. CPS enable this by continuously updating the digital representation as progressively more information about the physical entity becomes available. For example, when a patient experiences an elevated blood pressure, the CPS documents these changes and transmits the information to the digital twin, which may then simulate possible outcomes or trigger warnings. This facilitates pre-emptive actions, such as early detection of adverse events, automatic adjustment of doses, or reservation of the diagnostic follow-ups, and shifts the mode of response to care to a proactive one.[83–85]

5.3.3. *Secure communication*

CPS architecture is designed to use secure communication protocols

that ensure the safe transmission of sensitive health information. These systems must comply with strict healthcare regulations, such as Health Insurance Portability and Accountability Act (HIPAA) in the U.S. or the General Data Protection Regulation (GDPR) in the EU, which define how data should be stored, accessed, and shared. CPS systems involve strong cybersecurity measures, including firewalls, authentication protocols, and encrypted transmission channels, that help minimize the risks of unauthorized access or data leakage.[86]

5.3.4. Combinations of edge and cloud computing in CPS

The combination of cloud and edge computing in cyber-physical systems (CPS) enables efficient data processing by leveraging the strengths of both paradigms to meet the demands of healthcare applications. Cloud computing introduces scalability and computing power to analyze large and complex datasets and store long-term health information. The most important idea about edge computing is to move the computation nearer to the place where data is generated.[87]

For example, a medical device is located at the bedside of a patient or in an ambulance and performs low-latency, real-time analytics. This hybrid approach increases the responsiveness of the system, allows timely clinical decision-making, and prevents the centralized servers from being overloaded.[88]

5.3.5. Feedback loops practice

Among the typical attributes of advanced digital twin systems is the fact that they create closed-loop feedback. Knowledge generated in the virtual model determines the behavior of its physical counterpart.[12] CPS make this possible not only by receiving and sending data but also by accepting outputs and taking actions in the real world. For example, in a closed-loop insulin supply system for a diabetic patient, the digital twin will predict glucose spikes and instruct an associated insulin pump via the CPS to deliver the required amount of insulin.[89] The loop is an intelligent, self-adaptable system that enhances patient safety and treatment.[17] Cyber-Physical Systems are a critical part of making digital twins in healthcare successful.[90] They transform stand-alone datasets that are static into interrelated dynamic systems that simulate, predict, and optimize in the real world.[12] CPS permit real-time data interactions, ensure data security, enable rapid computing, and provide actionable feedback, making them central to DT technology in clinical practice. CPS will also become increasingly important in supporting precision medicine, operational optimization, and modern intelligent healthcare as the digital twin environment evolves.[90]

6. Enabling technologies

The digital twins in healthcare will be realized based on a complex of higher-end enabling technologies which simply join the physical and virtual worlds. These technologies can allow continuous capture, processing, and analysis of the high volumes of health-related data, and, thus, can allow digital twins to be a mirror of the current physiological and clinical situation of any patient, organs, or healthcare systems.[91] The underlying technology platform is Internet of Medical Things (IoMT), that offers a way to link diverse collections of medical devices, equipment, sensors and wearable digital technologies in order to accept continuous delivery of valuable medical patient data. They can serve as sensorial organs of the virtual twins that deliver real-time information to virtual models and are easy to track continuously and react promptly to patients' well-being changes. The development of Artificial Intelligence (AI) and Machine Learning (ML) is also an issue of equal concern. All these technologies allow digital twins to digest such dense data, recognize micro-patterns, and undertake prognostic research in order to assist with individual care and upfront care plans. AI and ML algorithms are used to process raw data to produce valuable information about the subsequent progress of the diseases, how it may be treated, and how medical treatment may be optimized. This heavy flow of data in terms of volume, variety and velocity can be supported with the help of Big Data analytics,[92] and[93,94] Cloud Computing, and processed with the assistance of a vast infrastructure. Cloud computing offers data storage and computing power, and the capacity to be increased according to the data size to be processed and the complexity of algorithms to be employed.

This infrastructure also provides collaborative research, sharing of information and real-time computing that is essential in the dynamism of the digital twins.[94]The third building block is 3D imaging and digital modelling that adds anatomical and functional refinements to organs and tissues. Such higher resolution models generate the structural basis of digital twins to be applied in surgical planning, diagnostics and biomechanical modelling. Last but not least, data standards and interoperability assure a high level of communication and data sharing among the other systems, devices and databases that are implemented into a digital twin environment.[95,96] Standardized data will enable cross-platform and institutional integration, enable data integrity preservation and coordinate multidisciplinary care. These empowering technologies combined form an effective framework that can transform the frozen health information into dynamics, interactive, and personalized digital twin models digitally.[97] They are the building blocks that justify that digital twins are the game changers to transforms healthcare delivery with the help of improved diagnosis and treatment, monitoring, and efficient operations.[96]

6.1. Internet of medical things (IoMT)

A bit narrower in scope to the healthcare industry, the Internet of Medical Things (IoMT) is simply a network of devices that are connected, analyze, collect, and transmit health- related data in real-time and, therefore, is an essential part of bringing digital twins to the healthcare sector. IoMT devices include wearable, implantable, remote monitoring, and connected medical devices to measure physiological and environmental data, which is increasingly being collected nonstop.[91,92] These devices will provide the stream of information necessary to develop and maintain a digital twin in which virtual models can reflect an accurate depiction of their real-to-life counterparts. One of the primary features of IoMT is the possibility to obtain the data in real-time.[98]

Glucose or intracranial pressure may be measured by implantable sensors and can be used in wearable devices (smartwatches) to measure heart rate, blood oxygen, or physical activity. This continuous observation is essential to patient-specific digital twins that strongly imply real-time input to simulate the existing health conditions, predict diseases and treatments, and customize a prescription.[99] Moreover, IoMT also enables remote monitoring in practicing medicine, significantly reducing the number of visits to hospitals. This kind of decentralization is especially significant in the management of chronic diseases when a living data stream is fed into the tuning of the digital twins to dynamic health conditions.[62] With IoMT, potential complications may be observed in time and inform the medical staff members themselves, as well as enable them to implement timely measures. The medical information should be treated as sensitive; hence, security and privacy are the main issues in IoMT.[100] The advanced encryption software, efficient security guidelines, and compliance to healthcare sector benchmarks (such as HIPAA - Health Insurance Portability and Accountability Act) implemented ensure that the medical information of the patients do not suffer leaks whatsoever and can never be subject to any cyberattacks.[79] The aspect of integration is a problem because of the heterogeneity of IoMT devices, various manufacturers, and standards of communication. However, the standardization and interoperability efforts are underway, which will make data flow easy and make digital twin systems more scalable.[97,101]

IoMT devices deliver real-time physiological data that is important in building DTs. Wearables, connected medical devices, and sensors monitor the heart rate, oxygen saturation, glucose levels, and body temperature constantly, inputting them into the DT system and dynamically updating the models of patients. In a 2025 survey, it was

found that DT frameworks vary in their levels of integration of IoMT: highly-integrated platforms are capable of near real-time monitoring and remote care, and low-integration platforms only offer offline simulation.[102] IoMT allows increasing patient engagement, facilitating telemedicine implementation, and lessening the necessity to visit the hospital frequently, which increases its access and compliance.[103]

6.2. Machine learning, artificial intelligence, big data and cloud computing

The core technologies are Artificial Intelligence (AI) and Machine Learning (ML), which transform raw data collected from patients and healthcare settings into useful insights in digital twins. These technologies can also be used to support advanced analytics, pattern recognition, and predictive modelling, which implies that digital twins are not only copies but dynamic systems that can learn as new data is continually introduced to them. Machine learning algorithms scan large amounts of data from electronic health records, genomic sequences, imaging studies, and wearable sensors to identify patterns and associations that may be hidden from human clinicians. This enables the evolution of diseases, response to treatment, and risk classification through predictive analytics. To illustrate the point, predicting how a patient will evolve over time or respond to various treatments can be conducted using ML models, enabling clinicians to implement interventions more accurately. A form of AI, deep learning, aims to support recognition and natural language processing and is therefore helpful in the understanding of medical imaging, pathology slides, and clinical notes. These capabilities, when incorporated into digital twins, increase the precision of organ-level or patient-specific models since comprehensive investigations of complex data types are displayed. In addition, digital twins can use AI-based simulation models in order to test hypotheticals (i.e., drug interactions or surgical outcomes) without endangering patient safety.[104]

Such a simulation potential allows making clinical decisions and individual treatment plans digital twins can be tailored in real-time with the application of reinforcement learning. In the actual world and forms its forecasting capacity in time. Anomaly detection is also aided by AI, where healthcare providers should detect some deviation in the expected pattern and consider this an indication of complications or emergency. Obstacles are the requirements of high amounts of quality data to be able to train models and the risk of bias in artificial intelligence algorithms, as well as the perception to seek explanations and to be able to interpret the AI models' decisions to gain clinician confidence. Regulatory deliberation is being changed to accommodate these concerns and to legitimize AI utilization in clinics.[55]

The fundamental aspect of DT frameworks is AI, which can be used to model predictions, simulate, and support decision making. It enables DTs to keep learning based on patient specific data and predicting the development of the disease, treatment results, and complication risk. Other uses of AI include state-of-the-art analytics of the system level, which can be either hospital resource demand prediction or patient flow prediction. Precision in intervention planning can be enhanced because the AI models are flexible to be customized to particular clinical areas, including cardiology, oncology, or diabetes.[105,106]

In a nutshell, AI and ML serve as the key enablers lifting digital twins to a new level of intelligent systems, which could provide decision support in real-time, personalized care, or continual learning. This is Big Data and Cloud Computing are key enablers in execution and scalability of digital twin in healthcare. Big data is the large and complicated information created by many digital sources of healthcare data, Such as electronic health records, medical imaging data, genomic information, wearable sensors, and patient-reported outcomes. This large volume of is manageable and extractable meaning. data require sophisticated storage, processing, and analysis system. The big data is managed in a safe environment based on cloud computing along with a promising and expandable infrastructure.[55] Cloud computing eliminates the need for physical hardware and high initial investments while enabling researchers and healthcare providers to perform powerful, on-demand computing. Such scalability is essential for precision in digital twin modelling, which requires continuous updating of large volumes of diverse data streams. Cloud infrastructures also facilitate the sharing of research and pooling of data across institutions, eliminating data silos that have historically hindered healthcare innovation types.[107] Cloud-based systems allow interoperability, enabling the integration of different kinds and formats of data into digital twin ecosystems.

Moreover, cloud computing supports sophisticated analytics, AI, real-time processing, and predictive modeling workloads.[94] This functionality is critical for applications in emergency planning, patient-specific monitoring, and treatment simulation that follows in Fig. 3. Some of the most critical aspects of deploying healthcare digital twins on the cloud are compliance and security. Rigorous cybersecurity measures and robust data governance systems are required to maintain patients' trust and ensure that confidential health information is neither misused nor exposed.[94] In essence, cloud computing, combined with Big Data, provides a foundation capable of meeting the data-intensive requirements of digital twins, enabling real-time analysis, collaboration, and scalable system-wide deployment in healthcare.[17,73,96,104,108] Cloud allows storage, processing and multi-center deployment of DT frameworks to be scaled. The reviews focus on stressing that cloud or edge computing is necessary to process large dimensional data of IoMT devices and to train AI models effectively.[105] Cloud-based DTs enable remote access to clinicians, enable multiple data sources to be integrated, and collaboration across institutions. Nonetheless, the implementation of clouds presents latency, cybersecurity threats, and regulatory compliance issues with governmental laws like the GDPR or the HIPAA. Edge computing may be used together to minimise latency and store sensitive data on the edge.

6.3. 3D imaging and digital modelling

The 3D imaging and digital modelling technologies are essential for creating detailed and precise digital twins, particularly at the organ and tissue level. Medical imaging modalities such as Magnetic Resonance Imaging (MRI), Computed Tomography (CT), Ultrasound, and Positron Emission Tomography (PET) provide high-resolution, three-dimensional views of body structures and physiological functions. This radiographic data is then translated into computerized models, which serve as the foundation for organ-specific digital twins. These images are reconstructed into volumetric and interactive models using advanced software that can be used by clinicians or researchers to assess structural integrity, functionality, and pathological changes. Surgical planning is more accurate through 3D digital models, which allow virtual rehearsal of complex procedures. Medical practitioners can navigate the patient's personal anatomy, anticipate potential issues, and simplify surgical plans, ultimately reducing operative risk and improving outcomes. In addition, biomechanical characteristics and physiological processes can be modeled. For example, blood flow dynamics in the cardiovascular system or airflow in the lungs. This enhances the predictive capacity of the digital twin by allowing scenario testing in addition to predicting physiological responses.[56,109] Digital modelling can, in some cases, be combined with 3D printing technologies to create physical models of organs or tumors that can be utilized in educational settings, in order to aid in the progress of the diagnosis, or to aid in preparations prior to an operation. These physical prototypes help bridge the gap between virtual simulations and the real world. The upgraded digital twins leverage the cross-indexing of medical images with datasets from numerous sources (genomics, biochemical, environmental, etc.), providing a more detailed account of patient health. Key challenges include the high computational demand required to process and render three-dimensional models, which must be precise and reproducible across all imaging modalities.[46,110,111]

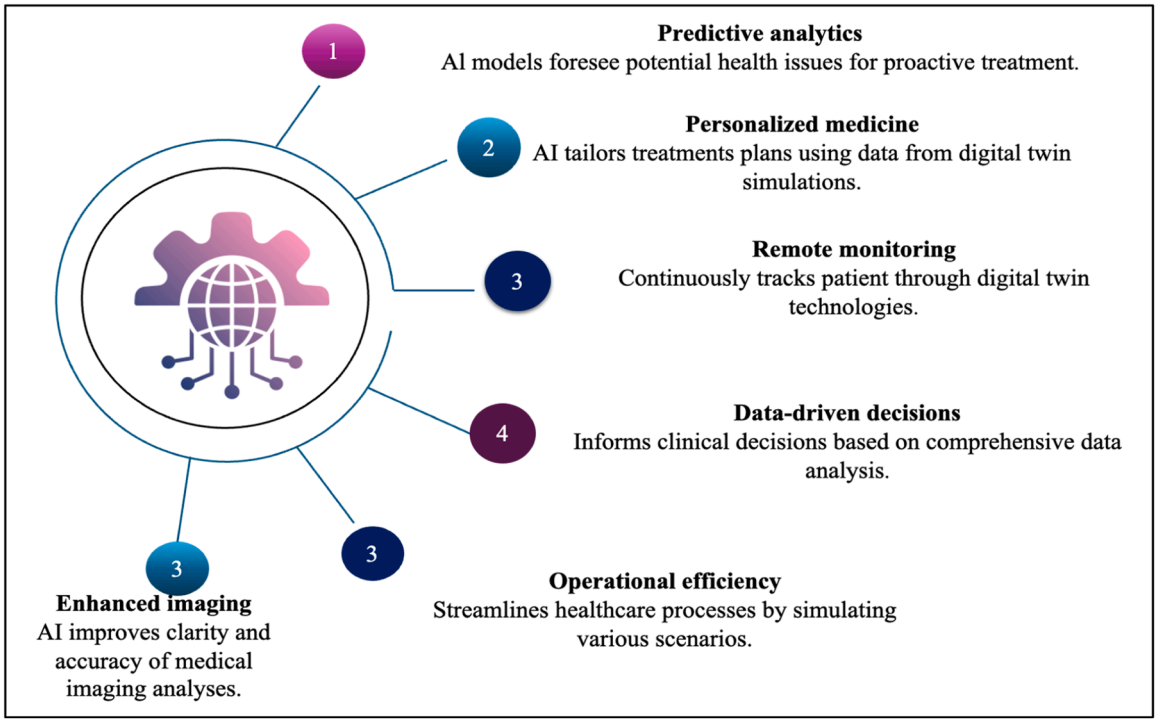


Fig. 3. Impact Artificial Intelligence (AI) and Machine Learning (ML) on Digital Twin.[55]

7. Interoperability and data standards

Interoperability, data protocols, and the effective use of digital twins in healthcare are founded on these pillars: A standardization of data. Interoperability can be understood as the relationships among information systems, devices, and applications that support access, exchange, and integration of data and its environment-wide cooperative use.[112] Medical data are heterogeneous in nature with a variety of sources, which include electronic health records (EHRs), imaging systems, laboratory instruments, wearables, and genomic databases. In the absence of interoperability, all these data sources will be in silos and are not useful to build complete digital twins.[44] The data communication standards (Standardized data formats and communication protocols such as HL7 (Health Level 7), FHIR (Fast Healthcare Interoperability Resources), DICOM (Digital Imaging and Communications in Medicine), and LOINC (Logical Observation Identifiers Names and Codes) can facilitate the free flow of information with ease. The standards will enable the accuracy of interpretation and use of data that has been recorded in one system by another without compromising data quality and the right context.[113] Interoperability enables the increasing scalability of digital twin systems since they would be able to connect new data streams and devices with no need to heavily customize the system.[114] It also facilitates collaborative care because multidisciplinary teams are in a position to share a single patient model. Nonetheless, there are still issues, namely, non-uniform adoption of the standards, the limits posed by legacy systems, and data privacy concerns. International and healthcare IT consortia work to standardize adoption across the board.[115] Moreover, policies like data governance, guiding and addressing topics of consent management, ownership and secure data sharing, play a key role in safe and compliant interoperability.[116]

8. Current application in healthcare

The use of digital twin technology is transforming healthcare by improving the way diseases are diagnosed, treated, and managed. It also helps hospitals operate more efficiently and supports medical research. By creating detailed virtual models of patients, organs, or entire healthcare systems, doctors and researchers can simulate outcomes, make predictions, and test treatment plans in real time. The main applications of this technology are outlined below.

8.1. *Personalized medicine individualized treatment planning*

Digital twins are highly personalized virtual models created using a wide range of patient data, including electronic health records, genomics, medical imaging, and information from wearable sensors. This comprehensive integration allows clinicians to construct accurate representations of a patient's anatomy, physiology, and disease state. Within these digital models, medical practitioners can simulate treatment strategies and test them virtually before applying them in real-world practice. For instance, in oncology, a digital twin of a cancer patient can be used to predict how tumors will respond to different chemotherapy drugs, enabling oncologists to design treatments that maximize efficacy while minimizing side effects.[5]

Furthermore, healthcare is shifting from a trial-and-error approach toward a more individualized, data-driven strategy. Digital twins enable the anticipation of individual patient behavior, helping to reduce adverse drug reactions, optimize dosing, and improve overall therapy success. This not only improves patient outcomes but also reduces healthcare expenditures by avoiding ineffective therapies.[117]

DTs combine AI, IoMT, and cloud computing to produce patient-specific virtual replicas, which allows simulating individual treatment plans. This enables clinicians to know how patients will respond to drugs, maximize efficacy and manage complications before they arise thus enhancing safety and effectiveness.[106] The personalized care based on genetic, lifestyle, and environmental factors allows the DTs to improve precision medicine, minimize adverse effects, and improve long-term health outcomes.

8.2. *Therapy outcomes-predictive modelling*

Digital twins enable the continuous monitoring and integration of new patient data, facilitating predictive analytics. They allow clinicians to model disease progression and patient-specific responses to

treatments over time, providing valuable predictions for clinical decision-making.[17] In cardiovascular care, for example, a digital twin can simulate how a patient's heart will respond to medication, helping to anticipate complications such as arrhythmias or heart failure exacerbations. This foresight enables early medical interventions and dynamic adjustments to treatment plans. Risk stratification can also be applied using predictive scores to identify patients at higher risk, allowing for targeted surveillance or preventive measures.[44] By forecasting therapy outcomes, digital twins support a proactive rather than reactive, approach to patient care, enhancing long-term clinical management and improving overall patient outcomes that follows in Fig. 4. Additionally, digital twins accelerate medical device development by complementing the traditional lifecycle of physical prototyping, animal testing, and clinical trials. Virtual simulations reduce costs and de-risk the development process, enabling faster and safer innovation in medical devices.[49]

The integration of real-time information of IoMT devices and AI models in DT models will allow to detect the decline of health early. The DTs were capable of detecting cardiac events or diabetic complications with an estimated real-time accuracy of 95.45, prior to the traditional clinical alert.[103] The predictive ability results in timely intervention, low rates of hospitalization and an enhanced patient prognosis. Constant surveillance also contributes to the dynamic risk classification of high-risks groups.

8.3. Simulation-based testing

Recorded twins can simulate the functional activity of the devices under different physiological contexts and personalize patient-specific variables. For example, an artificial heart valve digital twin can mimic the hemodynamics of blood flow and its interaction with tissues, allowing engineers to identify design flaws and adjust the model before prototype fabrication.[118] Device designs can integrate patient-specific anatomies, such as 3D scans of patient vessels, enabling stents or other implants to conform precisely to the patient's anatomy, thereby reducing risks such as restenosis. Twins are also capable of simulating the device lifecycle and predicting wear and maintenance requirements, resulting in greater efficacy and improved patient acceptability.[46]

8.4. Optimization of a healthcare system

Healthcare delivery is a highly complex system shaped by interactions among patients, providers, resources, and infrastructure. Digital twins, which serve as real-time virtual representations of healthcare systems, play a vital role in optimizing operations by enabling data-driven decisions that improve efficiency, reduce costs, and enhance care quality.[119] Resource allocation is one key application of digital twins. Digital twins can simulate patient occupancy, resource use, staff allocation, and machinery utilization. By forecasting demand trends, for example during flu seasons or pandemics, hospitals can proactively adjust capacity, redeploy staff, and ensure that critical resources are available when needed.[120] Workflow optimization is another important advantage. By visualizing the complete patient journey from check-in to discharge, digital twins can identify bottlenecks and inefficiencies in clinical

Workflows.[121] They allow administrators and providers to test virtual process improvements before implementing changes in practice. Examples include optimizing operating room schedules and improving pharmacy supply chains, which can reduce delays, decrease medical errors, and enhance overall efficiency through simulations.[47,122]

The DTs enable hospitals and healthcare systems to simulate workflows, resource distribution, and efficiency. Based on meta-review research, DT simulations have the potential to optimize a schedule, decrease wastage of resources, and lower expenses.[13] DTs facilitate the data-based management and policy-making by estimating the number of patients, the number of beds, and equipment utilization. Systemic optimization also provides the ability to plan the emergency situations, staffing arrangements, and infrastructure investments in the long-term.

8.5. Surgery simulation and education

Success in surgical procedures relies on elaborate pre-planning and practical execution. Both planning and execution are improved with the use of digital twins, which are patient-specific, hyper-realistic simulations. Digital twins enable surgical simulations in virtual reality (VR), augmented reality (AR), with or without haptics. Trainees can practice complex or infrequent cases repeatedly, receiving concurrent feedback that improves skill acquisition and confidence. Predictive maintenance of medical equipment is essential, as equipment failures can lead to delays and compromise patient safety.[73] Digital twins continuously monitor the conditions of medical devices and can predict potential failures in advance. Sensors transmit operational data to twin equipment, which identifies anomalies indicating wear or malfunction. For example, a twin MRI scanner can detect degradation of coils or cooling

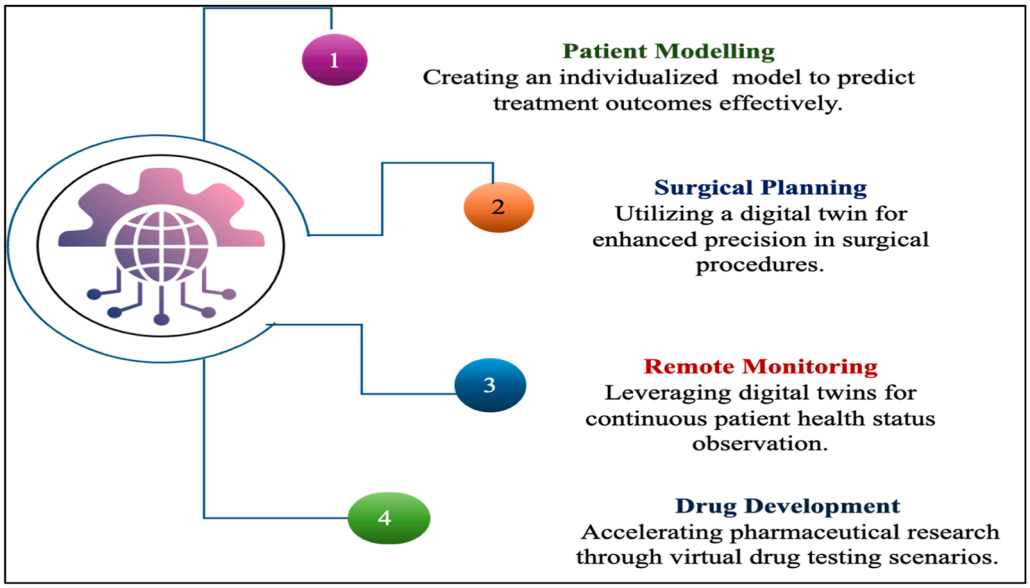


Fig. 4. Current applications of digital twin in healthcare.[17]

promptly. Predictive maintenance enabled by digital twins helps reduce equipment costs by prolonging lifespan, decreases emergency repairs, and prevents service interruptions, thereby enhancing patient safety.[73]

8.6. Continuous health monitoring

Patients continuously record vital signs, activity, sleep, and adherence data through their wearables and home devices, which continually contribute to their twins, alerting them when they are on the cusp of a health decline. Digital twins, for example, predict exacerbations by incorporating environmental factors, oxygen saturation, and respiratory rate in COPD. Twins create customized health alerts and advisories, such as cardiac arrhythmia forecasting, which help instruct clinicians and patients to make prompt care changes and increase both patient and clinician empowerment.[123,124]

8.7. Modelling of disease and epidemic control using digital twins

Digital twins can also be used with respect to public health, simulating populations at both the disease level and helping respond to epidemics. Anonymized data, aggregated by the twin's model, predicts disease patterns and all relevant influencing factors, thereby guiding targeted interventions. As an illustration, twins with diabetes forecast the subsequent burden of complications, which are used in screening and prevention. They also help epidemic management by modelling how infectious diseases spread and the effect of interventions (e.g., social distancing, vaccination), which helps inform adaptive, evidence-based policymaking, as seen in the context of COVID-19.[125,126]

8.8. Simulated, planned surgery and pre-operative digital questions

Surgical planning is one of the most promising applications of digital twins. Imaging data, such as CT or MRI scans, can be used to generate patient-specific virtual models of anatomical structures for surgeons. These carefully constructed virtual replicas enable surgeons to practice surgical procedures virtually and consider various surgical approaches, as well as predict vessel anomalies or tumor presence. Such digital rehearsal serves to minimize intraoperative surprises, reduce surgery time and the likelihood of complications. In cardiovascular surgery, surgeons may run a valve repair simulation on a patient's digital twin, practicing the procedure before the operation.[127]

Such practice is conducted prior to surgery and increases confidence and precision, resulting in improved patient outcomes that follows in Fig. 5. Surgeon and risk assessment training: Digital twins also offer risk assessment capabilities because they model various surgeries to measure potential undesirable events. They can identify hypothetical risky situations through such simulations and help derive mitigation strategies. In addition, digital twins are employed in medical and surgeon training using high-tech methods. Digital twins can assist in virtual reality (VR) scenarios that provide simulated, valuable experience without endangering patient safety. Trainees can repeat complex surgeries many times, enhancing skills acquisition and reducing the learning curve.[126,128]

8.9. Chronic real-time monitoring and intervention

Certain ailments, such as diabetes, heart failure, and chronic obstructive pulmonary disease (COPD), are chronic and that require continuous attention to prevent complications and emergency hospital admissions. Digital twins, combined with physiological information (including glucose levels) gathered from wearable devices and implantable sensors, are used to support ongoing monitoring.[129] Activity patterns, oxygen saturation, heart rate variability, and other levels are continuously monitored. This real-time data flows to the patient's digital twin, updating it as changes occur, so that deviations from normal health can trigger early warnings.[130] For example, a digital twin can predict the occurrence of diabetes in a patient, allowing insulin adjustments to be made before hyperglycemia develops.[125] Heart failure patients can undergo tests using digital twins to monitor fluid accumulations and cardiac performance, enabling early interventions before symptoms escalate.[131] Continuous monitoring and forecasting assist patients in self-care and help clinicians deliver prompt, individualized therapy. These approaches reduce emergency care and readmissions, thereby improving quality of life and conserving healthcare expenditure in hospital and clinical operations.[129]

8.10. Patient flow modelling

Digital twins are not limited to individual patients; they are also employed to model hospital systems and processes. By doing so, predicting bottlenecks and optimizing patient flows become possible. Emergency departments and intensive care units are examples of critical units simulated through patient admissions, transfers, and discharges.

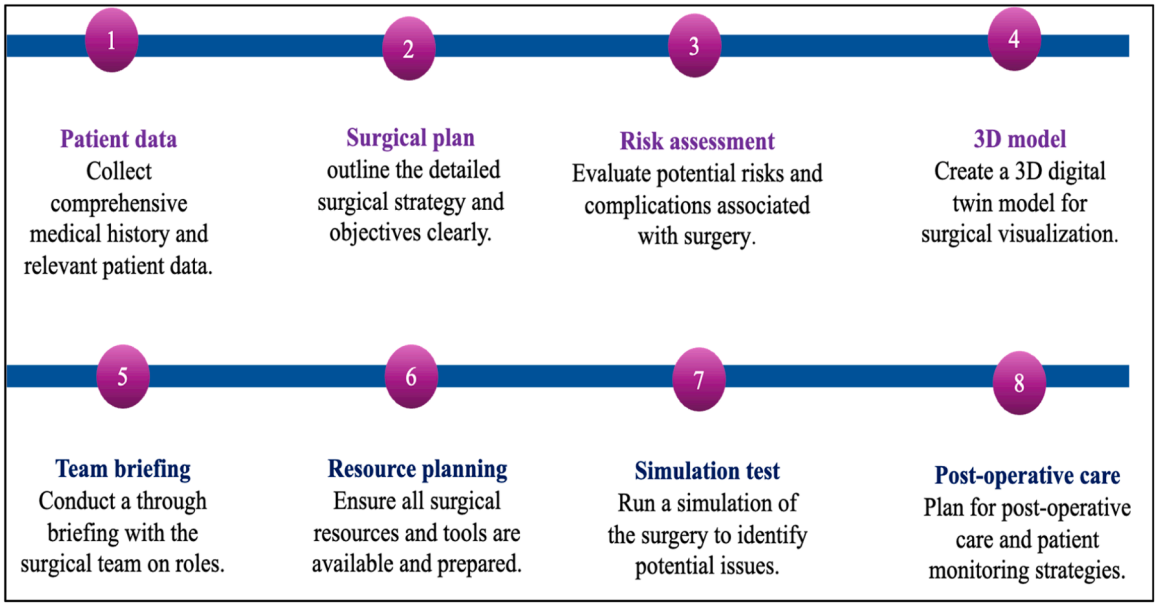


Fig. 5. Stimulated, Planned Surgery and Pre-Operative Digital Questions.[126,127]

These models enable administrators to make informed planning decisions,[132] considering periods of high demand, scarce resources, and patient wait times. One application is proactive capacity planning, predicting ICU bed usage, ventilator requirements, and staffing demands during ICU occupancy or a pandemic.

8.11. Resource and workflow optimization

Digital twins enable hospitals to model different staffing conditions, equipment availability, and procedures.[133] This allows for resource optimization, reduces operational costs, and improves the efficiency of care. Digital twins can also be used to design layouts that contain the spread of infections or develop optimized patient transportation routes.[46] Operational digital twin management enhances healthcare systems' resilience, improves patient satisfaction, and prevents delays. This is particularly critical in high-demand or crisis situations, where efficient system functioning can save lives.[134,135]

8.12. Pharmaceutical research and development

The process of developing drugs is usually costly and time-consuming. The use of digital twins is a promising way out, as it can be used to conduct in silico clinical trials in which virtual groups representing actual patient profiles are used to simulate drug reactions.[70] This enables researchers to conduct reviews on the safety and efficacy of new compounds across a variety of demographics and different genetic profiles. Virtual trials are able to identify potential drug candidate earlier, optimize dosage routes, and predict adverse reactions, thereby reducing the need for animal testing and early-phase human clinical trials. This shortens the drug development pipeline, allowing for faster production of medications.[136] Shrinkage of trial cost and duration: Through virtual simulations, pharmaceutical companies can save significant amount of money.[137] Digital twins reduce the need for large sample sizes, prolonged follow- up periods, and expensive site maintenance. Digital twin simulations are gaining recognition as credible evidence in regulatory approval processes, such as those overseen by the FDA.[150] Moreover, virtual trials support the advancement of precision medicine by identifying subgroups of patients who benefit most from a given treatment, improving trials success rate and fostering individualized therapeutic development.[46]

9. Benefits of digital twins in healthcare

Digital twin technology has the potential to create revolutionary improvements throughout healthcare, uniting real-time data, analytics, and simulation to foresee challenges, individualize care, and rationalize resources. Primary advantages:

A. **Patient outcomes:** Enhanced patient outcomes measure the general attitude of patients regarding the outpatient services provided. Personalized, predictive medicine will provide better health, reduce complications, and improve quality of life, including a personalized approach to medicine through improved treatments, detection, operational accuracy, and responsive adjustment of chronic diseases.[138]
B. **Reduced costs:** Digital twins allow significant cost savings without affecting the quality by preventing pointless treatments, accelerating device development, optimizing operations, extending device life, and reducing surgical complications.[139]
C. **Optimized care**: Digital twins can achieve real-time monitoring, effective team coordination, telehealth scale-up, patient engagement, and personalized rehabilitation, turning care into a proactive, continuous, patient-centered process.[103]
D. **Increased efficiency**: Automated data integration, workflow and staffing efficiencies, supply chain management, emergency preparation, and more are supported by health systems, allowing them to do more with less, reducing wait times, burnout, and wasteful operations.[139]
E. **Better decision-making**: Digital twins combine various patient and system data to support evidence-based clinical decision-making, risk stratification, strategic planning, personalized counselling, and continuous refinement of healthcare.[140]

10. Challenges and limitations

Digital twin technology holds remarkable promise in transforming healthcare delivery, personalized medicine, and clinical research. However, numerous challenges and limitations must be addressed to ensure its safe, effective, and ethical integration into healthcare systems. These issues span ethical, technical, regulatory, and infrastructural domains. Overcoming them is critical for realizing the full potential of digital twins while safeguarding patients and healthcare providers.[62]

10.1. Ethical and legal concerns (privacy, consent)

Digital twins need sustained access to large volumes of sensitive patient data, such as biometric data, medical history, lifestyle information and genetic profiles. This raises significant ethical issues concerning patient privacy and autonomy. Ensuring that patients provide informed, continuous consent for the collection, use, and dissemination of their data is a foundational ethical imperative. Unlike single-time data collection, digital twins are dynamically updated with new information, making it difficult to maintain ongoing consent over time. Patients may not be fully aware of all the applications or risks associated with their data being used in digital twin models.[141] Another issue is the control and ownership of data: who owns the digital twin or data derived from it, the patient, the healthcare professional, or the technology provider? Initial legal frameworks addressing these matters are incomplete or still developing in most jurisdictions. Liability also becomes complex when decisions based on a digital twin lead to unfavorable outcomes, making it difficult to assign responsibility between clinicians, developers, and institutions. Clearly, ethical guidelines, open data policies, governance frameworks, and patient-centered consent mechanisms are in high demand to balance empowering individuals and fostering trust.[141]

Some of the ethical issues are transparency, accountability, bias, fairness and trust of clinicians in the AI outputs. DTs are able to affect important treatment decisions, and interpretability shortage can decrease clinician self-confidence.[13,142] The AI bias in training information may result in the unfair treatment of different populations, which is why it is important to promote ethical regulation, audits, and ongoing surveillance.

10.2. Cybersecurity threat and data security

One of the most sensitive types of information in the healthcare sector is healthcare data in the digital world. The combination of digital twins with interconnected medical devices, cloud solutions, and IoMT ecosystems exposes healthcare systems to cybersecurity risks. These risks include unauthorized data access, theft, ransomware attacks, data manipulation, and system outages. Such incidents not only jeopardize patient privacy but may also pose a threat to life if digital twin-controlled clinical decisions are affected. Safeguarding information security must be multi-layered, including encryption schemes, strong user authentication, round-the-clock security surveillance, and incident response measures. Since digital twins are dynamic and operate in real-time, ensuring data integrity and the absence of tampering or other threats is a priority. Protecting against false information injection is paramount. In addition, communication channels must ensure secure data sharing between devices, cloud servers, and clinical users to prevent interception or cyberattacks. healthcare networks are further complicated by their size and complexity, making the adoption of standardized, up-to-date cybersecurity measures essential. If one area of

the system is compromised, the effects can cascade, impacting multiple services that rely on the digital twin ecosystem.[143,144]

DTs need extensive aggregation of sensitive health information, which makes them more susceptible to violations and abuse. Such threats to privacy as re-identification, unauthorized sharing, and cyberattacks are possible.[55,142] Such techniques as anonymization, differential privacy, and federated learning can be used to alleviate risks but can decrease model fidelity or predictive accuracy. The adoption requires compliance with the privacy regulations of HIPAA, GDPR, and other laws.

10.3. Technical limitations (model accuracy, computational costs)

A digital twin is sensitive to the quality, thoroughness, accuracy, and promptness of its underlying data and algorithms. Simulation of complicated biological systems, disease progression, drug interactions, and organ interactions necessitates high-quality computational methods and big data.[145] Nevertheless, existing models tend to be simplified quantitative representations due to missing biological information or computational limitations. High-fidelity digital twins can only be achieved by integrating heterogeneous data types, such as imaging, genomics, clinical labs, and sensor data, which may differ in resolution, quality, and format.[146] Incomplete or noisy data may reduce the reliability of the model. Real-time updates also require substantial computational power, both in processing and storage. High-performance computing infrastructures, frequently cloud-based, are required but expensive and may limit scalability, particularly in resource-constrained settings.[147] Additionally, the trade-off between model complexity and interpretability poses a challenge that follows in Fig. 6. To trust digital twin predictions, clinicians require understandable and well-explained models; however, highly complex machine learning models may function as black boxes. Overcoming these technical barriers will necessitate innovations in data integration, explainability of machine learning, and computational optimization.[148]

10.4. Lack of regulatory guidelines and clinical validation

Digital twins are new technologies in healthcare that integrate software, hardware, and AI-based statistics in an active system. Current regulatory models, intended to apply to traditional medical pharmaceuticals or devices, frequently do not suffice to determine the safety, efficacy, and reliability of these integrated systems.[62] Regulatory authorities like the FDA, EMA and others are actively deliberating new assessment pathways, but comprehensive guidelines are still under development. The widespread use of clinical approval is hampered by the absence of clear regulatory approval pathways.[149] The validation of digital twins in clinical trials and in real-world settings requires rigorous studies to ensure they enhance outcomes without causing unintended harm. Such studies must evaluate patient safety, accuracy, predictive value, and user acceptance.[150] There are also data standards to consider. Quality, model transparency, and risk management should be ensured. The lack of standardized protocols makes it difficult to compare digital twin solutions and impedes collaborative development.[151] Regulators, researchers, clinicians, and industry require coordinated efforts to develop robust frameworks that balance innovation with patient safety.[152,153]

Laws and regulations are not at par with DT technologies. The pit between the medical device and wellness tool frequently puts DTs in a challenging position regarding liability and approval channels.[154] There are other regulatory issues in cross-border data sharing, multi-center deployment, interoperability and continuous model verification. It is vital to tackle them to make DT adoption safe and ethical.

10.5. Interaction with the current electronic health records (EHRs)

Digital twins depend on sustained and complete patient information, much of which is stored within Electronic Health Records (EHRs). However, EHR systems tend to be disjointed, proprietary, and often lack interoperability. Variations in data formats, coding systems, and update frequencies can impede smooth integration with digital twins that follows in Fig. 7. This fragmentation can result in partial, delayed, or intermittent data streams, which compromise the real-time updating and accuracy of digital twins.[155–157]

10.6. Technical limitations (model accuracy, computational costs)

A digital twin is sensitive to the quality, thoroughness, accuracy, and promptness of its underlying data and algorithms. Simulation of complicated biological systems, disease progression, drug interactions, and organ interactions necessitates high-quality computational methods and big data.[146] Nevertheless, existing models tend to be simplified quantitative representations due to missing biological information or computational limitations. High-fidelity digital twins can only be achieved by integrating heterogeneous data types, such as imaging, genomics, clinical labs, and sensor data, which may differ in resolution, quality, and format.[146] Incomplete or noisy data may reduce the reliability of the model. Real-time updates also require substantial computational power, both in processing and storage.

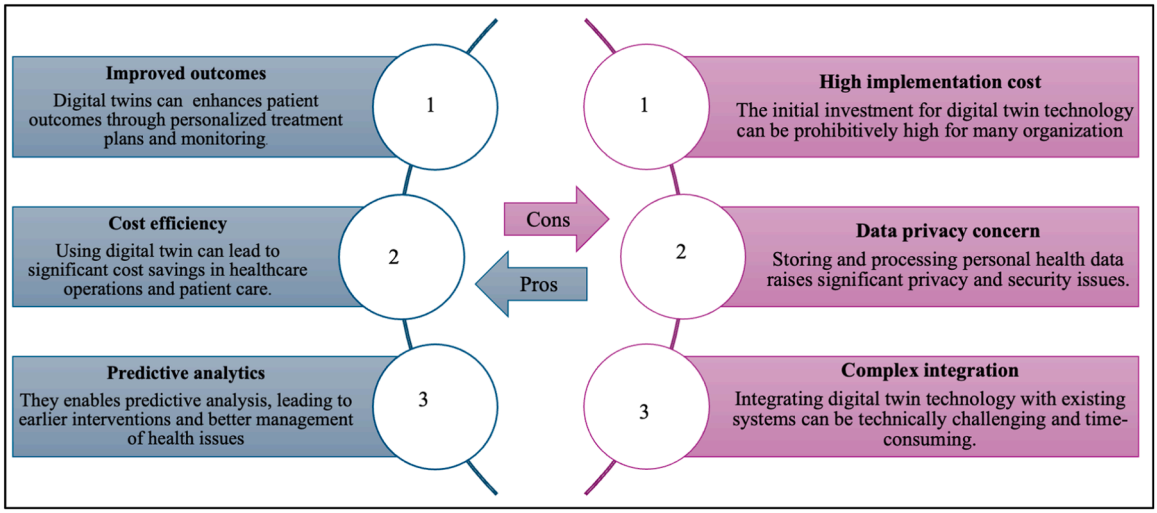


Fig. 6. Challenges and advantages of digital healthcare.[135]

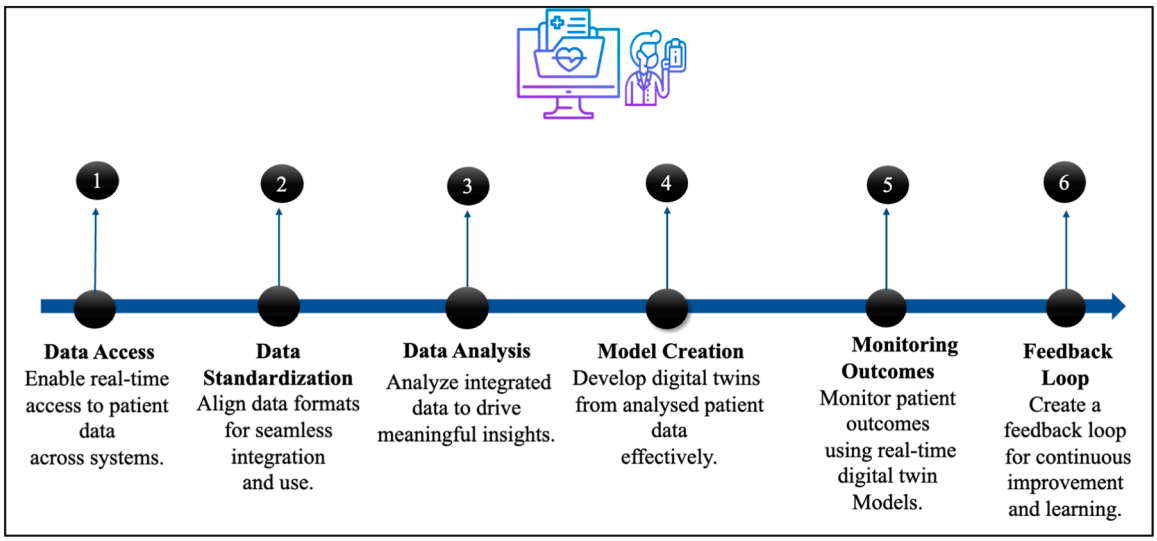


Fig. 7. Integrating with electronic health records.[158]

High-performance computing infrastructures, frequently cloud-based, are required but expensive and may limit scalability, particularly in resource-constrained settings.[147] Additionally, the trade-off between model complexity and interpretability poses a challenge. To trust digital twin predictions, clinicians require understandable and well-explained models; however, highly complex machine learning models may function as black boxes. Overcoming these technical barriers will necessitate innovations in data integration, explainability of machine learning, and computational optimization.[148]

11. Future opportunities and research directions

Digital twins represent a transformative frontier in healthcare, holding immense potential to reshape diagnosis, treatment, and healthcare delivery. As technology and science evolve, several exciting future opportunities and research directions are emerging that can further enhance the capabilities, scope, and impact of healthcare digital twins. These directions leverage advances in biology, artificial intelligence, computing, and health systems organization to expand the boundaries of what digital twins can achieve that follows in Fig. 8.

11.1. Integration of multi-omics data (genomics, proteomics, metabolomics)

One of the most promising frontiers for digital twins lies in the integration of multi-omics.

AI-driven digital twins could identify subtle signs of disease onset or progression well before clinical symptoms appear, allowing for earlier interventions. Predictive analytics within these digital twins might forecast outcomes such as treatment responses, risk of readmission, or adverse drug reactions with remarkable precision. By continuously learning from new patient data and trends in population health, these models can improve over time and adapt to emerging medical knowledge that follows in Fig. 9. Additionally, explainable AI techniques will be essential to ensure that clinicians can interpret and trust the recommendations.[159,160]

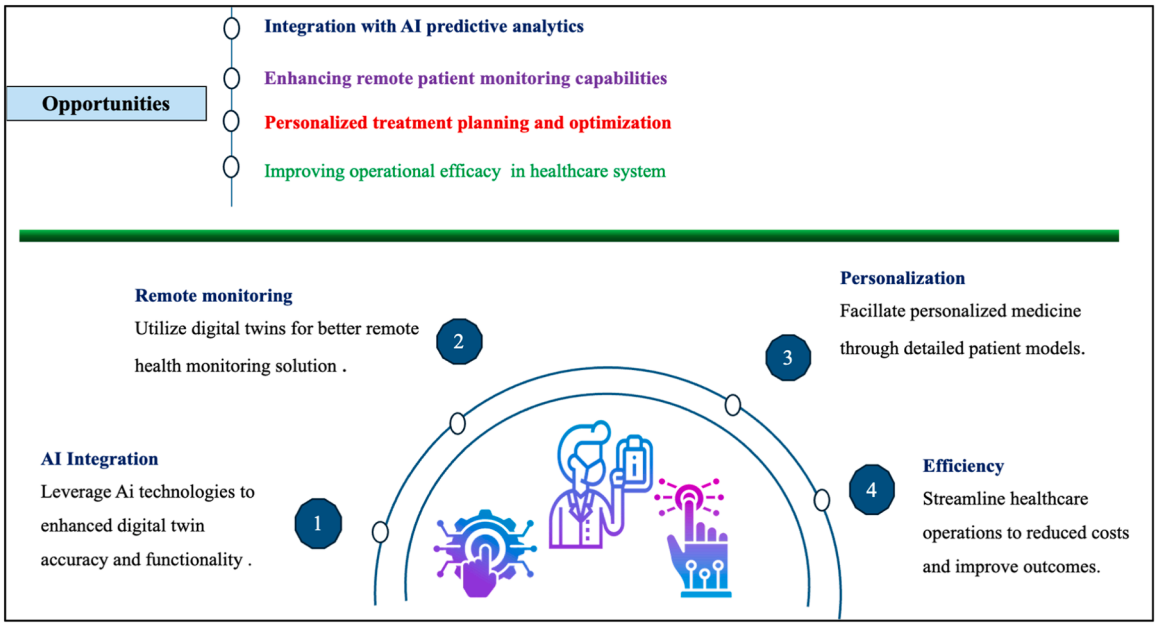


Fig. 8. Future Opportunities and Research Directions.[117,159,160]

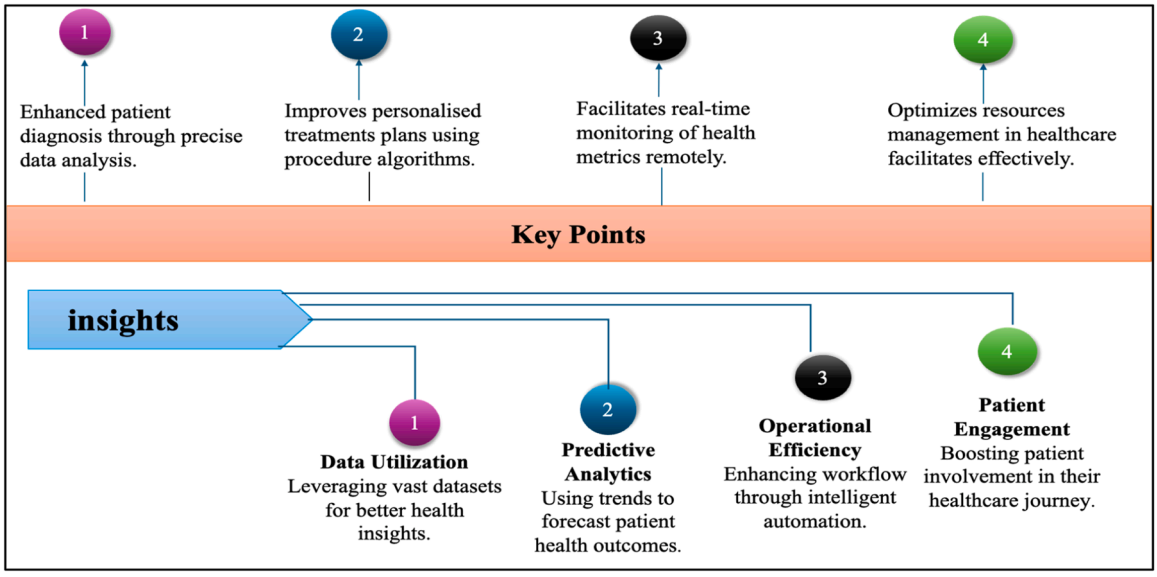


Fig. 9. Artificial Intelligence and Machine Learning Contribution.[159,160]

11.2. Nationwide/global digital twin frameworks

While individual and system-level digital twins are already emerging, the future could see the development of nationwide or even global digital twin frameworks. These large-scale systems would integrate digital twins across multiple healthcare providers, public health agencies, and research institutions to create comprehensive virtual replicas of population health.[53] Such frameworks would enable real-time epidemiological monitoring, early detection of outbreaks, and evaluation of public health interventions. By simulating population, dynamics, healthcare resource utilization, and social determinants of health, they have the potential to streamline regional, national, or international emergency preparedness and healthcare delivery. To realize this vision, common data-sharing protocols, cross-interoperability across varied healthcare IT systems, and strong privacy-assured data governance must be established. This also raises significant ethical challenges around surveillance and data sovereignty that must be addressed through open structures and inclusive participation.[122] Systems would integrate digital twins across multiple healthcare providers, public health agencies, and research institutions to create comprehensive virtual replicas of population health.[53] Such frameworks would enable real-time epidemiological monitoring, early detection of outbreaks, and evaluation of public health interventions. By simulating population, dynamics, healthcare resource utilization, and social determinants of health, they have the potential to streamline regional, national, or international emergency preparedness and healthcare delivery. To realize this vision, common data-sharing protocols, cross-interoperability across varied healthcare IT systems, and strong privacy-assured data governance must be established. This also raises significant ethical challenges around surveillance and data sovereignty that must be addressed through open structures and inclusive participation.[122]

11.3. Role in telehealth and remote monitoring

The COVID-19 pandemic stimulated the use of telehealth and remote patient monitoring- trends that are expected to continue growing. Digital twins could play a leading role in the development of this solution by constantly processing data from home monitoring devices, smartphones, and wearables. Through telehealth, clinicians would be able to interact with a patient's digital twin to detect chronic illnesses, adjust therapies, and intervene in a timely manner without the need for frequent hospitalization.

This potential would reduce healthcare costs, improve patient access in remote locations, and enhances patient engagement and self-management. Future studies will focus on improving the usability of digital twins for both patients and providers by integrating behavioral and environmental information to deliver holistic care, while ensuring secure and reliable data transmission. The fusion of digital twins with telehealth offers a powerful approach to proactive, continuous, and personalized healthcare delivery.[161] Ensuring safe and dependable data transfer is crucial. The combination of digital twins and telehealth provides a robust solution for proactive, ongoing, individualized healthcare. Digital twins in healthcare have a brighter future, full of transformative opportunities. Integrating multi-omics data will offer unprecedented accuracy in understanding and. treating diseases at the molecular level. Emergency and critical care real-time digital twins can save lives by providing fast and personalized treatments. AI-driven diagnostics and digital twins with predictive analytics will enhance early detection and individualized care plans. The integration of digital twins into national or international public health planning and response frameworks will transform how healthcare systems operate. Finally, the synergy of elevated accessibility to patient-centered telehealth and remote monitoring through digital twins will be further advanced. Resolving the technical, ethical, and infrastructural issues in these domains will be fundamental to achieving these opportunities and ushering in a new era of digital healthcare.[162]

12. Conclusion

Digital twins are changing the world of healthcare by creating dynamic virtual models that mirror real-time physical entities such as patients, organs, and health systems. This technology enables personalized care, improved surgical planning, chronic disease management, and streamlined hospital operations. Advances in IoMT, AI, big data, and 3D modeling have expedited their adoption, with promising results in improving patient outcomes and operational efficiency. Even though they hold promise, challenges such as data privacy, cybersecurity, model accuracy, and integration with existing systems remain significant obstacles. Overcoming these requires interdisciplinary cooperation among clinicians, engineers, data scientists, and policymakers. In addition, developing ethical principles and universally accepted protocols is challenging but essential to ensure patient safety, data security, and large-scale implementation. To conclude, digital twins have enormous

potential to transform healthcare by enabling precise, data-driven treatments and optimized clinical operations. With continued innovation, collaboration, and thoughtful regulation, digital twins can dramatically enhance healthcare quality, safety, and efficiency across the globe.

CRediT authorship contribution statement

Sakshi Parjane: Writing – review & editing, Methodology, Investigation. **Hrushikesh Mhaismale:** Writing – review & editing, Methodology, Investigation. **Gowtham Menon:** Writing – review & editing, Writing – original draft, Conceptualization. **Malave Bhagyashri Sunil:** Writing – original draft, Funding acquisition, Conceptualization. **Mayuri Mhaske:** Writing – original draft, Methodology, Investigation, Formal analysis.

Funding

Nil.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

All authors express gratitude to the Department of Science & Technology, Government of India, DIST-FIST 2015–2016 File No. SR/ FST/ college-272 dated 18 November 2015. New Delhi, for their financial support in providing the central instrumental facility at our institution, as well as acknowledging the organization of SRES, Sanjivani College of Pharmaceutical Education & Research (Autonomous), Kopargaon, India, for providing the necessary facilities for the completion of this research work.

References

1. Tao F, Zhang H, Liu A, Nee AYC. Digital twin in industry: state-of-the-art. *IEEE Trans Ind Inf*. 2019;15(4):2405–2415. https://doi.org/10.1109/TII.2018.2873186.
2. Bruynseels K, Santoni de Sio F, van den Hoven J. Digital twins in health care: ethical implications of an emerging engineering paradigm. *Front Genet*. 2018;9. https://doi.org/10.3389/fgene.2018.00031.
3. El Saddik A. Digital twins: the convergence of multimedia technologies. *IEEE Multimed*. 2018;25(2):87–92. https://doi.org/10.1109/MMUL.2018.023121167.
4. Kritzinger W, Karner M, Traar G, Henjes J, Sihn W. Digital twin in manufacturing: a categorical literature review and classification. *IFACPap*. 2018;51(11): 1016–1022. https://doi.org/10.1016/j.ifacol.2018.08.474.
5. Tao F, Cheng J, Qi Q, Zhang M, Zhang H, Sui F. Digital twin-driven product design, manufacturing and service with big data. *Int J Adv Manuf Technol*. 2018;94(9): 3563–3576. https://doi.org/10.1007/s00170-017-0233-1.
6. Chu CH, Wang L, Liu S, Zhang Y, Menozzi M. Augmented reality in smart manufacturing: Enabling collaboration between humans and artificial intelligence. *J Manuf Syst*. 2021;61:658–659. https://doi.org/10.1016/j.jmsy.2021.05.006.
7. Internet of Things with Deep Learning Techniques for Pandemic Detection: A Comprehensive Review of Current Trends and Open Issues. Accessed November 4, 2025. ⟨https://www.mdpi.com/2079-9292/13/13/2630⟩.
8. POTMEC: A Novel Power Optimization Technique for Mobile Edge Computing Networks. Accessed November 4, 2025. ⟨https://www.mdpi.com/2079-3197/13/7/161⟩.
9. Elgammal Z, Albrijawi MT, Alhajj R. Digital twins in healthcare: a review of AI-powered practical applications across health domains. *J Big Data*. 2025;12(1):234. https://doi.org/10.1186/s40537-025-01280-w.
10. Detecting latent topics and trends of digital twins in healthcare: A structural topic model-based systematic review - PubMed. Accessed November 4, 2025. ⟨https://pubmed.ncbi.nlm.nih.gov/37846404/⟩.
11. A Comprehensive Review on Cybersecurity of Digital Twins Issues, Challenges, and Future Research Directions | IEEE Journals & Magazine | IEEE Xplore. Accessed November 4, 2025. ⟨https://ieeexplore.ieee.org/abstract/document/10900372⟩.
12. Implementing Digital Twins That Learn: AI and Simulation Are at the Core. Accessed September 13, 2025. ⟨https://www.mdpi.com/2075-1702/11/4/425⟩.
13. Ringeval M, Etindele Sosso FA, Cousineau M, Paré G. Advancing Health Care With Digital Twins: Meta-Review of Applications and Implementation Challenges. *J Med Internet Res*. 2025;27, e69544. https://doi.org/10.2196/69544.
14. Halpern GA, Nemet M, Gowda DM, Kilickaya O, Lal A. Advances and utility of digital twins in critical care and acute care medicine: a narrative review. *J Yeungnam Med Sci*. 2025;42:9. https://doi.org/10.12701/jyms.2024.01053.
15. A Comprehensive Review on Digital Twin Applications in Healthcare System. ResearchGate. Accessed November 10, 2025. ⟨https://www.researchgate.net/publication/382648689_A_Comprehensive_Review_on_Digital_Twin_Applications_in_Healthcare_System⟩.
16. Machado TM, Berssaneti FT. Literature review of digital twin in healthcare. *Heliyon*. 2023;9(9), e19390. https://doi.org/10.1016/j.heliyon.2023.e19390.
17. Papachristou K, Katsakiori PF, Papadimitroulas P, Strigari L, Kagadis GC. Digital twins' advancements and applications in healthcare, towards precision medicine. *J Pers Med*. 2024;14(11):1101. https://doi.org/10.3390/jpm14111101.
18. Allen B.D. Digital Twins and Living Models at NASA. Presented at: Accessed September 13, 2025. ⟨https://ntrs.nasa.gov/citations/20210023699⟩.
19. The original twin. February 22, 2022. Accessed September 13, 2025. ⟨https://techinformed.com/the-original-twin/⟩.
20. Glaessgen E.H., Stargel D.S. The Digital Twin Paradigm for Future NASA and U.S. Air Force Vehicles. In: 2012. Accessed September 13, 2025. ⟨https://ntrs.nasa.gov/citations/20120008178⟩.
21. Apollo 13: The first digital twin - Simcenter. Accessed September 13, 2025. ⟨https://blogs.sw.siemens.com/simcenter/apollo-13-the-first-digital-twin/⟩.
22. Origins of the Digital Twin Concept. ResearchGate doi: 10.13140/RG.2.2.26367.61609.
23. Digital Twin: Mitigating Unpredictable, Undesirable Emergent Behavior in Complex Systems. Accessed September 13, 2025. ⟨https://www.researchgate.net/publication/306223791_Digital_Twin_Mitigating_Unpredictable_Undesirable_Emergent_Behavior_in_Complex_Systems⟩.
24. Digital Twins in HealthTech - Predictive Analytics & Personalized Care. Accessed September 13, 2025. ⟨https://www.augmentedcapital.co/blog/digital-twin-in-healthcare-a-game-changer-for-healthtech⟩.
25. Advancements in Digital Twin technology during the 2000s. DIGITAL TWIN. Accessed September 13, 2025. ⟨https://digitaltwintechnology.net/history/advancements-in-digital-twin-technology-during-the-2000s/?utm⟩.
26. Understanding the Role of Digital Twins in Manufacturing. Accessed September 13, 2025. ⟨https://aiiem.org/understanding-the-role-of-digital-twins-in-manufacturing/?utm⟩.
27. Digital Twin in Manufacturing: 10 Key Benefits & Use Cases. LITSLINK. July 29, 2024. Accessed September 13, 2025. ⟨https://litslink.com/blog/what-is-digital-twin-in-manufacturing-inspiring-industry-examples⟩.
28. Aircraft Engine Maintenance and Digital Twin Technology in Aircraft Engines. Accessed September 13, 2025. ⟨https://www.opastpublishers.com/open-access-articles/aircraft-engine-maintenance-and-digital-twin-technology-in-aircraft-engines-7660.html?utm⟩.
29. Jones D, Snider C, Nassehi A, Yon J, Hicks B. Characterising the digital twin: a systematic literature review. *CIRP J Manuf Sci Technol*. 2020;29:36–52. https://doi.org/10.1016/j.cirpj.2020.02.002.
30. Fuller A, Fan Z, Day C, Barlow C. Digital twin: enabling technologies, challenges and open research. *IEEE Access*. 2020;8:108952–108971. https://doi.org/10.1109/ACCESS.2020.2998358.
31. A Technological Review of Digital Twins and Artificial Intelligence for Personalized and Predictive Healthcare - PMC. Accessed September 13, 2025. ⟨https://pmc.ncbi.nlm.nih.gov/articles/PMC12294331/⟩.
32. Digital Twins Healthcare Impact, Adoption, Growth | Smart Spatial Blog. Accessed September 13, 2025. ⟨https://www.smartspatial.com/post/6-ways-digital-twins-in-healthcare-elevate-patient-outcomes-and-efficiency⟩.
33. Armeni P, Polat I, De Rossi LM, Diaferia L, Meregalli S, Gatti A. Digital twins in healthcare: is it the beginning of a new era of evidence-based medicine? A critical review. *J Pers Med*. 2022;12(8):1255. https://doi.org/10.3390/jpm12081255.
34. National Academies of Sciences E, Engineering NA of, Studies D on E and L, et al. *The digital twin landscape. Foundational Research Gaps and Future Directions for Digital Twins*. US: National Academies Press; 2024. ⟨https://www.ncbi.nlm.nih.gov/books/NBK605499/⟩. Accessed September 13, 2025.
35. The evolution of additive manufacturing in the aerospace industry. Surfeo. April 13, 2022. Accessed September 13, 2025. ⟨https://surfeo.eu/additive-manufacturing-the-evolution-of-in-the-aerospace-industry/⟩.
36. Digital twins in healthcare administration: A systematic review on novel approaches to management and planning | Russian Open Medical Journal. Accessed September 13, 2025. ⟨https://www.romj.org/2025-0104⟩.
37. Digital Twins Market Size: Comprehensive Growth Analysis 2024. Accessed September 13, 2025. ⟨https://www.marketsandmarkets.com/blog/SE/digital-twins-market-size-comprehensive-growth-analysis-2024⟩.
38. Revolutionizing Healthcare with Precision Medicine through Virtual Twins | Dassault Systèmes. Accessed September 13, 2025. ⟨https://www.3ds.com/insights/corporate-reports/transforming-medicine-virtual-twins-precision-healthcare⟩.
39. Accessed September 13, 2025. ⟨https://pmc.ncbi.nlm.nih.gov/articles/PMC12294331/32⟩.
40. Exploring approaches for predictive cancer patient digital twins: Opportunities for collaboration and innovation - PMC. Accessed September 13, 2025. ⟨https://pmc.ncbi.nlm.nih.gov/articles/PMC9586248/⟩.
41. Li H, Zhang J, Zhang N, Zhu B. Advancing Emergency Care With Digital Twins. *JMIR Aging*. 2025;8, e71777. https://doi.org/10.2196/71777.
42. Katsoulakis E, Wang Q, Wu H, et al. Digital twins for health: a scoping review. *NPJ Digit Med*. 2024;7:77. https://doi.org/10.1038/s41746-024-01073-0.

43. Human Digital Twins in Personalized Healthcare: An Overview and Future Perspectives. ResearchGate doi: 10.48550/arXiv.2503.11944.
44. Vallée A. Digital twin for healthcare systems. *Front Digit Health*. 2023;5, 1253050. https://doi.org/10.3389/fdgth.2023.1253050.
45. Zhang K, Zhou HY, Baptista-Hon DT, et al. Concepts and applications of digital twins in healthcare and medicine. *Patterns*. 2024;5(8), 101028. https://doi.org/10.1016/j.patter.2024.101028.
46. Kabir MR, Shishir FS, Shomaji S, Ray S. Digital twins in healthcare IoT: a systematic review. *HighConfid Comput*. 2025;5(3), 100340. https://doi.org/10.1016/j.hcc.2025.100340.
47. How Digital Twins In Healthcare Are Transforming Patient Care. ScalaCode. May 24, 2025. Accessed September 13, 2025. ⟨https://www.scalacode.com/blog/digital-twins-in-healthcare/⟩.
48. Batko K, Ślęzak A. The use of big data analytics in healthcare. *J Big Data*. 2022;9(1):3. https://doi.org/10.1186/s40537-021-00553-4.
49. Digital twins for health: a scoping review | npj Digital Medicine. Accessed September 13, 2025. ⟨https://www.nature.com/articles/s41746-024-01073-0⟩.
50. Digital twin for healthcare systems - PMC. Accessed September 13, 2025. ⟨https://pmc.ncbi.nlm.nih.gov/articles/PMC10513171/⟩.
51. Haleem A, Javaid M, Pratap Singh R, Suman R. Exploring the revolution in healthcare systems through the applications of digital twin technology. *Biomed Technol*. 2023;4:28–38. https://doi.org/10.1016/j.bmt.2023.02.001.
52. Digital Twins for Clinical and Operational Decision-Making: Scoping Review - PMC. Accessed September 13, 2025. ⟨https://pmc.ncbi.nlm.nih.gov/articles/PMC11754991/⟩.
53. Attaran M, Celik BG. Digital Twin: Benefits, use cases, challenges, and opportunities. *Decis Anal J*. 2023;6, 100165. https://doi.org/10.1016/j.dajour.2023.100165.
54. Liu X, Jiang D, Tao B, et al. A systematic review of digital twin about physical entities, virtual models, twin data, and applications. *Adv Eng Inf*. 2023;55, 101876. https://doi.org/10.1016/j.aei.2023.101876.
55. Saratkar SY, Langote M, Kumar P, Gote P, Weerarathna IN, Mishra GV. Digital twin for personalized medicine development. *Front Digit Health*. 2025;7, 1583466. https://doi.org/10.3389/fdgth.2025.1583466.
56. Zhao F, Wu Y, Hu M, et al. Current progress of digital twin construction using medical imaging. *J Appl Clin Med Phys*. 2025;26(9), e70226. https://doi.org/10.1002/acm2.70226.
57. Digital twins for cardiac electrophysiology: state of the art and future challenges - PMC. Accessed September 13, 2025. ⟨https://pmc.ncbi.nlm.nih.gov/articles/PMC11161534/⟩.
58. Fekonja LS, Schenk R, Schröder E, Tomasello R, Tomšič S, Picht T. The digital twin in neuroscience: from theory to tailored therapy. *Front Neurosci*. 2024;18, 1454856. https://doi.org/10.3389/fnins.2024.1454856.
59. HospiT'Win: A Predictive Simulation-Based Digital Twin for Patients Pathways in Hospital | Request PDF. In: ResearchGate. doi:10.1109/BHI.2019.8834534.
60. Cheng W, Lian W, Tian J. Building the hospital intelligent twins for all-scenario intelligence health care. *Digit Health*. 2022;8, 20552076221107894. https://doi.org/10.1177/20552076221107894.
61. Lin X, Kundu L, Dick C, Obiodu E, Mostak T. *arXiv*. Preprint posted online. *6G Digit Twin Netw Theory Pract*. December 5, 2022. https://doi.org/10.48550/arXiv.2212.02032.
62. Chaparro-Cárdenas SL, Ramirez-Bautista JA, Terven J, et al. A technological review of digital twins and artificial intelligence for personalized and predictive healthcare. *Healthcare*. 2025;13(14):1763. https://doi.org/10.3390/healthcare13141763.
63. Strocchi M, Hammersley DJ, Halliday BP, Prasad SK, Niederer SA. Cardiac digital twins: a tool to investigate the function and treatment of the diabetic heart. *Cardiovasc Diabetol*. 2025;24:293. https://doi.org/10.1186/s12933-025-02839-w.
64. Hughes A, Shandhi MMH, Master H, Dunn J, Brittain E. Wearable devices in cardiovascular medicine. *Circ Res*. 2023;132(5):652–670. https://doi.org/10.1161/CIRCRESAHA.122.322389.
65. Abadi E, Segars WP, Tsui BMW, et al. Virtual clinical trials in medical imaging: a review. *J Med Imaging*. 2020;7(4), 042805. https://doi.org/10.1117/1.JMI.7.4.042805.
66. Journal of Medical Internet Research - Digital Twins for Clinical and Operational Decision-Making: Scoping Review. Accessed September 13, 2025. ⟨https://www.jmir.org/2025/1/e55015/⟩.
67. Leviathan-like concerns and reflections on medical digital twin technology | AI & SOCIETY. Accessed September 13, 2025. ⟨https://link.springer.com/article/10.1007/s00146-025-02510-8⟩.
68. Fischer RP, Volpert A, Antonino P, Ahrens TD. Digital patient twins for personalized therapeutics and pharmaceutical manufacturing. *Front Digit Health*. 2024;5, 1302338. https://doi.org/10.3389/fdgth.2023.1302338.
69. Digital twins in oncology. Accessed September 13, 2025. ⟨https://www.researchgate.net/publication/368573527_Digital_twins_in_oncology⟩.
70. Pammi M, Shah PS, Yang LK, Hagan J, Aghaeepour N, Neu J. Digital twins, synthetic patient data, and in-silico trials: can they empower paediatric clinical trials? *Lancet Digit Health*. 2025;7(5), 100851. https://doi.org/10.1016/j.landig.2025.01.007.
71. Drummond D, Gonsard A. Definitions and characteristics of patient digital twins being developed for clinical use: scoping review. *J Med Internet Res*. 2024;26, e58504. https://doi.org/10.2196/58504.
72. Accessed September 13, 2025. ⟨https://www.researchgate.net/figure/The-digital-twin-service-management-and-applications_fig2_326029013⟩.
73. Mekki YM, Luijten G, Hagert E, et al. Digital twins for the era of personalized surgery. *NPJ Digit Med*. 2025;8:283. https://doi.org/10.1038/s41746-025-01575-5.
74. Pilosof NP, Barrett M, Oborn E, Barkai G, Zimlichman E, Segal G. Designing for flexibility in hybrid care services: lessons learned from a pilot in an internal medicine unit. *Front Med Technol*. 2023;5, 1223002. https://doi.org/10.3389/fmedt.2023.1223002.
75. Real-World Examples of Digital Twin Technology in Healthcare. December 20, 2023. Accessed September 13, 2025. ⟨https://arkenea.com/blog/digital-twin-technology-in-healthcare/⟩.
76. Enhancing Hospital Efficiency through IoT and AI: A Smart Healthcare System. Accessed September 13, 2025. ⟨https://www.researchgate.net/publication/384012959_Enhancing_Hospital_Efficiency_through_IoT_and_AI_A_Smart_Healthcare_System⟩.
77. Agius S, Magri C, Cassar V. A cognitive task analysis for developing a clinical decision support system for emergency triage. *J Emerg Nurs*. 2025. https://doi.org/10.1016/j.jen.2025.05.013.
78. Digital twin-driven intelligence disaster prevention and mitigation for infrastructure: advances, challenges, and opportunities. ResearchGate. Published online April 11, 2025. doi:10.1007/s11069-021-05190-x.
79. Vidyalakshmi G, Gopikrishnan S, Boulila W, Koubaa A, Srivastava G. Digital twins and cyber-physical systems: a new frontier in computer modeling. *Comput Model Eng Sci*. 2025;143(1):51–113. https://doi.org/10.32604/cmes.2025.057788.
80. A review of digital twins and their application in cybersecurity based on artificial intelligence | Artificial Intelligence Review. Accessed September 13, 2025. ⟨https://link.springer.com/article/10.1007/s10462-024-10805-3⟩.
81. Real-Time Patient Monitoring with IoT-Integrated Wearables. Accessed September 13, 2025. ⟨https://www.researchgate.net/publication/388892225_Real-Time_Patient_Monitoring_with_IoT-Integrated_Wearables⟩.
82. Human Digital Twin: Data, Models, Applications, and Challenges. Accessed September 13, 2025. https://arxiv.org/html/2508.13138v1.
83. Park D, So K, Prabhakar SK, et al. Early warning score and feasible complementary approach using artificial intelligence-based bio-signal monitoring system: a review. *Biomed Eng Lett*. 2025;15(4):717–734. https://doi.org/10.1007/s13534-025-00486-4.
84. Jiang Y, Li M, Li M, et al. Digital twin-enabled real-time synchronization for planning, scheduling, and execution in precast on-site assembly. *Autom Constr*. 2022;141, 104397. https://doi.org/10.1016/j.autcon.2022.104397.
85. Ooka T. The era of preemptive medicine: developing medical digital twins through Omics, IoT, and AI integration. *JMA J*. 2025;8(1):1–10. https://doi.org/10.31662/jmaj.2024-0213.
86. Digital Twins for Cyber-Physical Healthcare Systems: Architecture, Requirements, Systematic Analysis and Future Prospects. Accessed September 13, 2025. ⟨https://www.researchgate.net/publication/389618888_Digital_Twins_for_Cyber-Physical_Healthcare_Systems_Architecture_Requirements_Systematic_Analysis_and_Future_Prospects⟩.
87. Indonet. Hybrid Network: Combination of Cloud and Edge Computing. indonet.id. August 19, 2025. Accessed September 13, 2025. ⟨https://indonet.id/hybrid-network-combination-of-cloud-and-edge-computing/⟩.
88. Medical 4.0 technologies for healthcare: Features, capabilities, and applications - ScienceDirect. Accessed September 13, 2025. ⟨https://www.sciencedirect.com/science/article/pii/S2667345222000104⟩.
89. Templer S. Closed-loop insulin delivery systems: past, present, and future directions. *Front Endocrinol*. 2022;13, 919942. https://doi.org/10.3389/fendo.2022.919942.
90. The Role of Digital Twins in Collaborative Cyber-Physical Systems. Accessed September 13, 2025. ⟨https://www.researchgate.net/publication/340989243_The_Role_of_Digital_Twins_in_Collaborative_Cyber-Physical_Systems⟩.
91. Digital Twin: Enabling Technologies, Challenges and Open Research | IEEE Journals & Magazine | IEEE Xplore. Accessed September 13, 2025. ⟨https://ieeexplore.ieee.org/document/9103025⟩.
92. Analytics and Freemium Products - ScienceDirect. Accessed September 13, 2025. ⟨https://www.sciencedirect.com/science/article/abs/pii/B978012416966500002 6⟩.
93. A Review of the State of the Art for the Internet of Medical Things. Accessed September 13, 2025. ⟨https://www.mdpi.com/2413-4155/7/2/36⟩.
94. Paek SW, Lee Y, Park H, Shin H, Lee SW. Development of distributed network-based digital twin with real-time unstructured data processing: a case study on directed energy deposition system. *Manuf Lett*. 2025;44:1198–1208. https://doi.org/10.1016/j.mfglet.2025.06.139.
95. Digital Twins in Healthcare: Revolutionizing Patient Care at Duke - Center for Computational and Digital Health Innovation. Accessed September 13, 2025. ⟨https://comphealth.duke.edu/digital-twins-in-healthcare-revolutionizing-patient-care-at-duke/⟩.
96. 3D anatomical digital twins: New generation virtual models to navigate robotic partial nephrectomy. Accessed September 13, 2025. ⟨https://www.researchgate.net/publication/389074650_3D_anatomical_digital_twins_New_generation_virtual_models_to_navigate_robotic_partial_nephrectomy⟩.
97. Interactive Viral Marketing Through Big Data Analytics, Influencer Networks, AI Integration, and Ethical Dimensions. Accessed September 13, 2025. ⟨https://www.mdpi.com/0718-1876/20/2/115⟩.
98. Internet of Medical Things (IoMT)-Based Smart Healthcare System: Trends and Progress - PMC. Accessed September 13, 2025. ⟨https://pmc.ncbi.nlm.nih.gov/articles/PMC9308524/⟩.

99. Internet of robotic things for independent living: Critical analysis and future directions - ScienceDirect. Accessed September 13, 2025. ⟨https://www.sciencedirect.com/science/article/pii/S2542660524000623⟩.
100. Manickam P, Mariappan SA, Murugesan SM, et al. Artificial intelligence (AI) and internet of medical things (IoMT) assisted biomedical systems for intelligent healthcare. *Biosensors*. 2022;12(8):562. https://doi.org/10.3390/bios12080562.
101. Data Interoperability: Principles, Challenges, Best Practices. Accessed September 13, 2025. ⟨https://www.acceldata.io/blog/data-interoperability-key-principles-challenges-and-best-practices⟩.
102. Elgammal Z, Albrijawi MT, Alhajj R. Digital twins in healthcare: a review of AI-powered practical applications across health domains. *J Big Data*. 2025;12(1):234. https://doi.org/10.1186/s40537-025-01280-w.
103. Jameil AK, Al-Raweshidy H. A digital twin framework for real-time healthcare monitoring: leveraging AI and secure systems for enhanced patient outcomes. *Discov Internet Things*. 2025;5(1):37. https://doi.org/10.1007/s43926-025-00135-3.
104. Łukaniszyn M, Majka Ł, Grochowicz B, Mikołajewski D, Kawala-Sterniuk A. Digital twins generated by artificial intelligence in personalized healthcare. *Appl Sci*. 2024;14(20):9404. https://doi.org/10.3390/app14209404.
105. Pellegrino G, Gervasi M, Angelelli M, Corallo A. A conceptual framework for digital twin in healthcare: evidence from a systematic meta-review. *Inf Syst Front*. 2025;27(1):7–32. https://doi.org/10.1007/s10796-024-10536-4.
106. Chaparro-Cárdenas SL, Ramirez-Bautista JA, Terven J, et al. A technological review of digital twins and artificial intelligence for personalized and predictive healthcare. *Healthcare*. 2025;13(14):1763. https://doi.org/10.3390/healthcare13141763.
107. Deep learning for power quality - ScienceDirect. Accessed September 13, 2025. ⟨https://www.sciencedirect.com/science/article/pii/S0378779622009385⟩.
108. Pinto-Coelho L. How artificial intelligence is shaping medical imaging technology: a survey of innovations and applications. *Bioengineering*. 2023;10(12):1435. https://doi.org/10.3390/bioengineering10121435.
109. Portnoy Y, Koren J, Khoury A, et al. Three-dimensional technologies in presurgical planning of bone surgeries: current evidence and future perspectives. *Int J Surg Lond Engl*. 2023;109(1):3–10. https://doi.org/10.1097/JS9.0000000000000201.
110. Meyer-Szary J, Luis MS, Mikulski S, et al. The role of 3D printing in planning complex medical procedures and training of medical professionals—cross-sectional multispecialty review. *Int J Environ Res Public Health*. 2022;19(6):3331. https://doi.org/10.3390/ijerph19063331.
111. Digital Twin: Values, Challenges and Enablers From a Modeling Perspective. Accessed September 14, 2025. ⟨https://www.researchgate.net/publication/338869735_Digital_Twin_Values_Challenges_and_Enablers_From_a_Modeling_Perspective⟩.
112. Klonoff DC, Aaron RE, Tian T, Espinoza JC. Principles of Interoperability of diabetes devices. Published online August 5 *J Diabetes Sci Technol*. 2024. https://doi.org/10.1177/19322968241268235.
113. Dalal S, Jain S. Chapter 11 - Smart mental healthcare systems. In: Jain S, Jain V, Balas VE, eds. *Web Semantics*. Academic Press; 2021:153–163. https://doi.org/10.1016/B978-0-12-822468-7.00010-9.
114. Ehrenstein V, Kharrazi H, Lehmann H, Taylor CO. *Obtaining data from electronic health records. Tools and Technologies for Registry Interoperability, Registries for Evaluating Patient Outcomes: A User's Guide, 3rd Edition, Addendum 2 [Internet]*. Agency for Healthcare Research and Quality (US); 2019. ⟨https://www.ncbi.nlm.nih.gov/books/NBK551878/⟩.
115. Burns DJP, Arora J, Okunade O, et al. International consortium for health outcomes measurement (ICHOM): standardized patient-centered outcomes measurement set for heart failure patients. *Jacc Heart Fail*. 2020;8(3):212–222. https://doi.org/10.1016/j.jchf.2019.09.007.
116. Joyce A, Javidroozi V. Smart city development: data sharing vs. data protection legislations. *Cities*. 2024;148, 104859. https://doi.org/10.1016/j.cities.2024.104859.
117. Frontiers | Digital twin for personalized medicine development. Accessed September 14, 2025. ⟨https://www.frontiersin.org/journals/digital-health/articles/10.3389/fdgth.2025.1583466/full⟩.
118. Coorey G, Figtree GA, Fletcher DF, et al. The health digital twin to tackle cardiovascular disease—a review of an emerging interdisciplinary field. *NPJ Digit Med*. 2022;5:126. https://doi.org/10.1038/s41746-022-00640-7.
119. Sun Z, Jansen S. Personalized 3D printed coronary models in coronary stenting. *Quant Imaging Med Surg*. 2019;9(8):1356–1367. https://doi.org/10.21037/qims.2019.06.21.
120. Tyagi AK, Richa. Chapter 3 - digital twin–based healthcare facilities management. In: Nguyen TA, ed. *Blockchain and Digital Twin for Smart Hospitals*. 2025:47–65. https://doi.org/10.1016/B978-0-443-34226-4.00004-6. Elsevier.
121. Leveraging Digital Twins in Healthcare: Predictive Analytics for Patient Journey Optimization. Accessed September 14, 2025. https://www.researchgate.net/publication/392020761_Leveraging_Digital_Twins_in_Healthcare_Predictive_Analytics_for_Patient_Journey_Optimization.
122. Rehan MW, Rehan MM. Survey, taxonomy, and emerging paradigms of societal digital twins for public health preparedness. *NPJ Digit Med*. 2025;8:520. https://doi.org/10.1038/s41746-025-01737-5.
123. Clark M, Bailey S. Single-use wearable wireless sensors for vital sign monitoring: CADTH horizon scan. *Can Agency Drugs Technol Health*; 2023. ⟨http://www.ncbi.nlm.nih.gov/books/NBK599967/⟩.
124. Ghadi YY, Shah SFA, Waheed W, et al. Integration of wearable technology and artificial intelligence in digital health for remote patient care. *J Cloud Comput*. 2025;14(1):1–25. https://doi.org/10.1186/s13677-025-00759-4.
125. The potential of the Medical Digital Twin in diabetes management: a review - PMC. Accessed September 14, 2025. ⟨https://pmc.ncbi.nlm.nih.gov/articles/PMC10397506/⟩.
126. Digital-twin-based multi-scale simulation supports urban emergency management: a case study of urban epidemic transmission. Accessed September 14, 2025. ⟨https://www.tandfonline.com/doi/full/10.1080/17538947.2024.2421950⟩.
127. Asciak L, Kyeremeh J, Luo X, et al. Digital twin assisted surgery, concept, opportunities, and challenges. *NPJ Digit Med*. 2025;8:32. https://doi.org/10.1038/s41746-024-01413-0.
128. Ntakakis G, Plomariti C, Frantzidis C, Antoniou PE, Bamidis PD, Tsoulfas G. Exploring the use of virtual reality in surgical education. *World J Transpl*. 2023;13(2):36–43. https://doi.org/10.5500/wjt.v13.i2.36.
129. Long H, Li S, Chen Y. Digital health in chronic obstructive pulmonary disease. *Chronic Dis Transl Med*. 2023;9(2):90–103. https://doi.org/10.1002/cdt3.68.
130. Johnson Z, Saikia MJ. Digital twins for healthcare using wearables. *Bioengineering*. 2024;11(6):606. https://doi.org/10.3390/bioengineering11060606.
131. Trayanova NA, Prakosa A. Up digital and personal: how heart digital twins can transform heart patient care. *Heart Rhythm*. 2024;21(1):89–99. https://doi.org/10.1016/j.hrthm.2023.10.019.
132. Advances and utility of digital twins in critical care and acute care medicine: a narrative review. Accessed September 14, 2025. ⟨https://www.e-jyms.org/journal/view.php?number=2867⟩.
133. Unlocking the potential of digital twins: Transforming hospital practices for better care - J Community Health Manag. Accessed September 14, 2025. ⟨https://jchm.in/archive/volume/11/issue/2/article/14186#article⟩.
134. Kuruppu Appuhamilage GDK, Hussain M, Zaman M, Ali Khan W. A health digital twin framework for discrete event simulation based optimised critical care workflows. *NPJ Digit Med*. 2025;8:376. https://doi.org/10.1038/s41746-025-01738-4.
135. DelveInsight. Digital Twins Applications and Challenges in the Healthcare Domain. DelveInsight Business Research. January 31, 2024. Accessed September 14, 2025. ⟨https://www.delveinsight.com/blog/digital-twin-technology-challenges-and-applications⟩.
136. Exploring the Potential of Digital Twins in Cancer Treatment: A Narrative Review of Reviews. Accessed September 14, 2025. ⟨https://www.mdpi.com/2077-0383/14/10/3574⟩.
137. How Digital Twins Transform Drug Development Processes. Accessed September 14, 2025. ⟨https://www.biospectrumindia.com/views/17/26602/how-digital-twins-transform-drug-development-processes.html⟩.
138. Gibbons C, Porter I, Gonçalves-Bradley DC, et al. Routine provision of feedback from patient-reported outcome measurements to healthcare providers and patients in clinical practice. *Cochrane Database Syst Rev*. 2021;10(10), CD011589. https://doi.org/10.1002/14651858.CD011589.pub2.
139. The Value of Digital Twin Technology. Accessed September 14, 2025. ⟨https://healthmanagement.org/c/hospital/whitepaper/the-value-of-digital-twin-technology⟩.
140. Pash G., Villa U., I.I. DAH, Yankeelov T.E., Willcox K. Predictive Digital Twins with Quantified Uncertainty for Patient-Specific Decision Making in Oncology. arXiv. Preprint posted online May 13, 2025. doi:10.48550/arXiv.2505.08927.
141. Ethical and legal challenges with IoT in home digital twins - ScienceDirect. Accessed September 14, 2025. ⟨https://www.sciencedirect.com/science/article/pii/S2215016125002559⟩.
142. Bäumer FS, Schultenkämper S, Geierhos M, Lee YS. Mirroring privacy risks with digital twins: when pieces of personal data suddenly fit together. *SN Comput Sci*. 2024;5(8):1109. https://doi.org/10.1007/s42979-024-03413-z.
143. Pärn E, Ghadiminia N, García de Soto B, Oti-Sarpong K. A perfect storm: digital twins, cybersecurity, and general contracting firms. *Dev Built Environ*. 2024;18, 100466. https://doi.org/10.1016/j.dibe.2024.100466.
144. Cybersecurity challenges of digital twins: threats and security measures | INCIBE-CERT | INCIBE. Accessed September 14, 2025. ⟨https://www.incibe.es/en/incibe-cert/blog/cybersecurity-challenges-digital-twins-threats-and-security-measures⟩.
145. Challenges and opportunities for digital twins in precision medicine from a complex systems perspective | npj Digital Medicine. Accessed September 14, 2025. ⟨https://www.nature.com/articles/s41746-024-01402-3⟩.
146. Pros and Cons of Digital Twin Technology. Accessed September 14, 2025. ⟨https://www.belcan.com/2023/01/16/pros-and-cons-of-digital-twin-technology/⟩.
147. Advantages and disadvantages of digital twin technology | TechTarget. Search ERP. Accessed September 14, 2025. ⟨https://www.techtarget.com/searcherp/feature/Advantages-and-disadvantages-of-digital-twin-technology⟩.
148. Digital Twin Architecture: Pros & Cons Explained. Accessed September 14, 2025. ⟨https://www.toobler.com/blog/digital-twin-architecture-pros-cons⟩.
149. (PDF) Digital twins and AI for end-to-end sustainable pharmaceutical supply chain management. Accessed September 14, 2025. ⟨https://www.researchgate.net/publication/390456622_Digital_twins_and_AI_for_end-to-end_sustainable_pharmaceutical_supply_chain_management⟩.
150. The status quo and future prospects of digital twins for healthcare - ScienceDirect. Accessed September 14, 2025. ⟨https://www.sciencedirect.com/science/article/pii/S2950489924000423⟩.
151. The Applications and Challenges of Digital Twin Technology in Smart Grids: A Comprehensive Review. Accessed September 14, 2025. ⟨https://www.mdpi.com/2076-3417/14/23/10933⟩.
152. Borkar S., Chakole S., Prasad R., Bansod S. Revolutionizing Oncology: A Comprehensive Review of Digital Health Applications. *Cureus*. 16(4):e59203. doi: 10.7759/cureus.59203.
153. Macdonald JC, Isom DC, Evans DD, Page KJ. Digital innovation in medicinal product regulatory submission, review, and approvals to create a dynamic

regulatory ecosystem—are we ready for a revolution? *Front Med*. 2021;8, 660808. https://doi.org/10.3389/fmed.2021.660808.
154. Delerm F, Pilottin A. Double edged tech: navigating the public health and legal challenges of digital twin technology. *Eur J Public Health*. 2024;34(Supplement_3), ckae144.1510. https://doi.org/10.1093/eurpub/ckae144.1510.
155. Electronic Health Records (EHRs): Opportunities and Challenges in Data Sharing and Interoperability. Accessed September 14, 2025. ⟨https://www.researchgate.net/publication/390877736_Electronic_Health_Records_EHRs_Opportunities_and_Challenges_in_Data_Sharing_and_Interoperability⟩.
156. Why Is the Electronic Health Record So Challenging for Research and Clinical Care? - PMC. Accessed September 14, 2025. ⟨https://pmc.ncbi.nlm.nih.gov/articles/PMC9295893/⟩.
157. Common EHR Implementation Challenges and Solutions. Accessed September 14, 2025. ⟨https://www.narolainfotech.com/blogs/ehr-implementation-challenges-and-solutions/⟩.
158. Alkan M., Huijsdens H., Jones Y., Deligianni F. Electronic Health Records: Towards Digital Twins in Healthcare. arXiv.org. January 16, 2025. Accessed September 14, 2025. ⟨https://arxiv.org/abs/2501.09640v2⟩.
159. AI-Driven Predictive Analytics, Healthcare Outcomes, Cost reduction, Machine Learning, Patient Monitoring. Accessed September 14, 2025. ⟨https://www.researchgate.net/publication/388402701_AI-Driven_Predictive_Analytics_Healthcare_Outcomes_Cost_reduction_Machine_Learning_Patient_Monitoring⟩.
160. AI-Powered Digital Twins in Healthcare: Predictive Modeling for Personalized Treatment and Surgery Planning. Accessed September 14, 2025. ⟨https://www.researchgate.net/publication/390761322_AI-Powered_Digital_Twins_in_Healthcare_Predictive_Modeling_for_Personalized_Treatment_and_Surgery_Planning⟩.
161. Vallée A. Envisioning the future of personalized medicine: role and realities of digital twins. *J Med Internet Res*. 2024;26, e50204. https://doi.org/10.2196/50204.
162. Digital Twin in Healthcare - Transforming Patient Care. Accessed September 14, 2025. ⟨https://emorphis.health/blogs/digital-twins-in-healthcare/⟩.